EXACT CRITERION FOR A SUM OF TWO HIGHER CLOSED-NULL SETS TO COVER HIGHER CANTOR SPACE

ADAM MARTON, JAROSLAV SUPINA, MIROSLAV REPICKY, TRISTAN VAN DER VLUGT

Editorial note. Complete original mathematical body and original bibliography retained; author responsibility for mathematics, no independent certification, repaired proof or added assumptions.

Editorial note. Native disclosure400 and footnote926 retained: two additive-failure lemma statements were strengthened while testing ChatGPT5.6, but authors explicitly say all proofs are their own. This is not converted into a core-proof AI-discovery claim or a source-wide AI-free claim.

Editorial note. Unavailable Biber setup replaced by natbib/BibTeX with native main.bib fields; derived types thesis/online use BibTeX phdthesis/misc, math-title case protected and dash/apostrophe encoded. Raw source/bib preserved.

Editorial note. Equivalent clauses, additive-failure support lemmas and direct corollaries are not separate counts. All global cardinal hypotheses and theorem-specific stronger assumptions retained.

Editorial note. Original source references the unresolved label thm:Ekm not ideal. Raw source is preserved; the rendered reference is explicitly marked unresolved without assigning a substitute target.

Allowed standard prerequisites (editorial).

- Higher Cantor space, regular-cardinal partitions, slaloms and cardinal characteristics. Native global regular-uncountable and kappa=kappa^{<kappa} conventions, stronger limit/inaccessible hypotheses where used, block/interval partitions, higher ideal definitions and complete supporting arguments retained.. Entire original mathematical body, original TikZ diagrams and complete main.bib supplied; all distinctions among E/BE and existential/universal generators retained.

1. Introduction

In the last two decades, a considerable effort has gone into the generalisation of combinatorics of the Baire space ${}^{\omega}\omega$ to the context of the *higher* Baire space ${}^{\kappa}\kappa$, where κ is regular uncountable. Although the higher context resembles the classical context closely in many cases, it can behave very differently in others: statements that are consistent in the classical context (e.g., $\mathfrak{b} < \mathfrak{s}$), turn out to have inconsistent higher analogues ($\mathfrak{s}_\kappa \leq \mathfrak{b}_\kappa$ is a theorem, [24]), or vice versa.

In some cases, the classically defined object does not have a clear generalisation. This is the case for Lebesgue measure, which is, due to its σ-additivity, heavily dependent on the separability of the reals, cf. [1]. For this reason, it is hard, or even impossible, to say what the generalisation of a Lebesgue null set should be (although several attempts have been made, see [29, 14, 12]). Nevertheless, through classically equivalent combinatorial notions, it is possible to generalise stronger concepts. For instance, even without Lebesgue measure, it is not hard to generalise the notion of strongly null, also known as strong measure zero, cf. [18, 19].

Date: arxiv:2610.05784v1; CC BY 4.0.

In this article we will take the classical σ-ideal $\mathcal{E}$ that is generated by the *closed Lebesgue null* sets of the Cantor space ${}^{\omega}2$, and generalise it to the context of the higher Cantor space ${}^{\kappa}2$. We do so by carefully avoiding Lebesgue measure — instead, we use a characterisation of closed null sets due to Bartoszyński and Shelah [4], which is based on slaloms and interval partitions. This allows us to introduce two κ-complete ideals, $\mathcal{E}_\kappa$ and $\mathcal{BE}_\kappa$, that are equivalent in the classical context, but distinct in the higher context.

1.1. **Overview.** In Section 2, we introduce candidates for a higher closed null ideal and study their combinatorial properties. We summarise the classical definition and combinatorial characterisation of the closed null ideal $\mathcal{E}$ on the Cantor space (Section 2.1), before we define the higher analogues $\mathcal{E}_\kappa$ and $\mathcal{BE}_\kappa$ and study their basic properties (Section 2.2). We use a family of parametrised generating ideals $\mathcal{E}_\kappa(B)$ for this definiton, which also provides us with the families $\mathcal{E}_\kappa^\exists$, $\mathcal{E}_\kappa^\forall$, $\mathcal{BE}_\kappa^\exists$ and $\mathcal{BE}_\kappa^\forall$ consisting of unions or intersections of such generating ideals (Definition 2.7). We compare $\mathcal{E}_\kappa$ and $\mathcal{BE}_\kappa$ to the κ-meagre ideal $\mathcal{M}_\kappa$ (Section 2.3), and we investigate the relations between individual generating ideals based on their parameters (Section 2.4).

Section 3 concerns the closure of the higher closed null ideals under binary addition. After we introduce the necessary definitions and introduce the additive ideals $\mathcal{NA}_\kappa$, $\mathcal{MA}_\kappa$ and the strongly null ideal $\mathcal{SN}_\kappa$ (Section 3.1), we study the relation between these ideals, the higher closed null additive ideals and the ideals $\mathcal{E}_\kappa^\forall$ and $\mathcal{BE}_\kappa^\forall$ (Section 3.2). We finish the section by considering failures of additivity, where adding together two small families results in a large sum (Section 3.3)

Finally, Section 4 analyses the cardinal characteristics of these ideals, and compares them with each other (Section 4.1) and with some other cardinal characteristics (Section 4.2), and Section 5 looks at the effect of some forcing notions on the values of these cardinal characteristics.

1.2. **Preliminaries.** We will always assume that κ is a regular uncountable cardinal with $\kappa = \kappa^{<\kappa}$, but often we have to make stronger assumptions, e.g. that κ is regular limit or strongly inaccessible. Let $\langle X_\alpha \mid \alpha \in \kappa\rangle$ be a sequence of sets such that $2 \leq |X_\alpha| \leq \kappa$ for all α, and let $\mathcal{X} = \prod_{\alpha\in\kappa} X_\alpha$. We give $\mathcal{X}$ a topology generated by the base $\big\{[s] \;\big|\; s \in \bigcup_{\gamma<\kappa} \prod_{\alpha\in\gamma} X_\alpha\big\}$, where we define $[s] = \{f \in \mathcal{X} \mid s \subseteq f\}$. We will refer to such spaces $\mathcal{X}$ simply as *higher product spaces*. Among the most well-studied such spaces are the *higher Baire space* ${}^{\kappa}\kappa$ and the *higher Cantor space* ${}^{\kappa}2$, in analogy with the *classical Baire space* ${}^{\omega}\omega$ and *classical Cantor space* ${}^{\omega}2$.

We say a set $X \subseteq \mathcal{X}$ is *κ-meagre* if it is the union of κ-many nowhere dense sets, where a set Y is *nowhere dense* if every basic set $[s]$ contains some basic set $[t]$ with $Y \cap [t] = \varnothing$. We let $\mathcal{NWD}(\mathcal{X})$ be the family of nowhere dense subsets of $\mathcal{X}$ and $\mathcal{M}_\kappa(\mathcal{X})$ be the family of κ-meagre subsets of $\mathcal{X}$. It is not hard to see that $\mathcal{NWD}(\mathcal{X})$ is a $<\kappa$-complete proper ideal on $\mathcal{X}$ and $\mathcal{M}_\kappa(\mathcal{X})$ is a $\leq\kappa$-complete proper ideal on $\mathcal{X}$. Here we call an ideal *$<\kappa$-complete*, if it is closed under unions of size $<\kappa$, and *$\leq\kappa$-complete* if it is $<\kappa^+$-complete. We will write $\mathcal{NWD}$ and $\mathcal{M}_\kappa$ if the space $\mathcal{X}$ is clear from context.

A property holds for *almost all* $\alpha \in \kappa$ if there is $\alpha_0 \in \kappa$ such that it holds for all $\alpha \geq \alpha_0$. Similarly, a property holds for *cofinally many* $\alpha \in \kappa$ if for each $\alpha_0 \in \kappa$ there is $\alpha \geq \alpha_0$ for which it holds. We abbreviate these as the quantifiers $\forall^\infty\alpha \in \kappa$ and $\exists^\infty\alpha \in \kappa$, respectively. We write $[X]^\kappa$ for the set of subsets of X of size κ, and similarly we write $[X]^{<\kappa}$ for $\bigcup_{\lambda<\kappa}[X]^\lambda$. Two sets $X, Y \in [\kappa]^\kappa$ are called *almost disjoint* if $|X \cap Y| < \kappa$, and X is called an *almost subset* of Y, denoted $X \subseteq^* Y$ if X and $\kappa \setminus Y$ are almost disjoint, or equivalently if $|X \setminus Y| < \kappa$.

1.3. **Acknowledgements.** No LLMs were used in the writing and research of this article, except for Lemmas 2.27 and 2.29, which were strengthened from an earlier form to their current statement while testing ChatGPT 5.6. All proofs are our own.

2. Generalising the closed null ideal

The σ-ideal $\mathcal{E}$ generated by closed sets of Lebesgue measure zero (in the space ${}^{\omega}2$) was given a combinatorial characterisation in terms of slaloms by Bartoszyński and Shelah [4, Theorem 4.3]. We will repeat this characterisation in Section 2.1, providing the classical framework with which the generalised context could be compared. For detailed information about the classical ideal $\mathcal{E}$ we refer to [3, § 2.6], or recently [10].

Lebesgue measure does not appear in the abovementioned combinatorial characterisation of $\mathcal{E}$. This allows us to define a higher closed null ideal for uncountable regular cardinals κ. We provide two different approaches in Section 2.2 and compare the resulting ideals $\mathcal{E}_\kappa$ and $\mathcal{BE}_\kappa$ to each other and to the κ-meagre ideal $\mathcal{M}_\kappa$. These three ideals turn out to be comparable yet distinct, or to be precise $\mathcal{E}_\kappa \subsetneq \mathcal{BE}_\kappa \subsetneq \mathcal{M}_\kappa$.

2.1. **The classical closed null ideal.** The ideal $\mathcal{E}$ is the σ-ideal generated by the family of all closed subsets of the Cantor space ${}^{\omega}2$ that have Lebesgue measure zero, where we recall that the Lebesgue measure μ on ${}^{\omega}2$ is the standard product measure, where we endow $2 = \{0,1\}$ with the *coin-flip measure* $\mu_{1/2}$, that is, $\mu_{1/2}(\{0,1\}) = 1$ and $\mu_{1/2}(\{0\}) = \mu_{1/2}(\{1\}) = \frac{1}{2}$. It is well-known that the theory of Lebesgue measure cannot be easily generalised to the higher context, and recent work (cf. [1, 33]) has shown that any such generalisation has to be severly limited. We will therefore use a different definition of $\mathcal{E}$ that is classically equivalent. Our presentation of this characterisation is the one provided in [10, Theorem 2.3], which itself is a structural refinement of the characterisation provided in [4, Theorem 4.3].

Let ℓ^1_+ be space of sequences $\varepsilon\colon \omega \to (0,\infty)$ of positive real numbers, such that $\sum_{n\in\omega}\varepsilon_n$ converges. Given such $\varepsilon \in \ell^1_+$, let us for convenience define the multiplicative inverse $h_\varepsilon\colon n \mapsto \frac{1}{\varepsilon_n}$. An *interval partition* of ω is a sequence $I = \langle I_n \mid n\in\omega\rangle$ of mutually disjoint nonempty intervals of natural numbers such that $\omega = \bigcup_{n\in\omega} I_n$. We will adopt the convention that $\min(I_{n+1}) = \max(I_n)+1$ for all $n\in\omega$, which also implies $\min(I_0) = 0$. We denote by IP the family of all such interval partitions on ω. We define a family of slaloms parametrised by $\varepsilon\in\ell^1_+$ and $I\in\mathrm{IP}$ (cf. [10, Definition 2.1]), and rewrite it in order to make it more easily comparable to our definition for the uncountable case:

$$\begin{aligned}
\Sigma(I,\varepsilon) &= \left\{\varphi\in\textstyle\prod_{n\in\omega}\mathcal{P}\left({}^{I_n}2\right) \;\middle|\; \lim_{n\to\infty}\frac{|\varphi(n)|}{2^{|I_n|}\cdot\varepsilon_n} = 0\right\}\\
&= \left\{\varphi\in\textstyle\prod_{n\in\omega}\mathcal{P}\left({}^{I_n}2\right) \;\middle|\; \forall N\in\omega\,\forall^\infty n\in\omega\left(\frac{|\varphi(n)|}{2^{|I_n|}} < \frac{\varepsilon_n}{N}\right)\right\}\\
(*)\qquad &= \left\{\varphi\in\textstyle\prod_{n\in\omega}\mathcal{P}\left({}^{I_n}2\right) \;\middle|\; \forall N\in\omega\,\forall^\infty n\in\omega\left(N\cdot h_\varepsilon(n)\cdot|\varphi(n)| < 2^{|I_n|}\right)\right\}.
\end{aligned}$$

We may now define for a given slalom $\varphi\in\Sigma(I,\varepsilon)$ the following families:

$$\begin{aligned}
[\varphi]_* &= \{x\in{}^{\omega}2 \mid \forall^\infty n\in\omega\,(x\restriction I_n\in\varphi(n))\},\\
\mathcal{E}(I,\varepsilon) &= \{X\subseteq{}^{\omega}2 \mid \exists\varphi\in\Sigma\,(I,\varepsilon)(X\subseteq[\varphi]_*)\}.
\end{aligned}$$

We are now ready to give the combinatorial characterisation of $\mathcal{E}$.

Theorem 2.1 ([10, Theorem 2.3])**.** $\mathcal{E} = \bigcup_{I\in\mathrm{IP},\,\varepsilon\in\ell^1_+}\mathcal{E}(I,\varepsilon)$.

Let us take a moment to see how we may generalise this characterisation to uncountable κ. The use of converging series of real numbers has no clear analogue, but our inverse function h_ε provides a solution to this problem; the intuitive idea is that a series $\sum\varepsilon_n$ is convergent if h_ε grows fast enough. For instance, if $\varepsilon_n = n^{-r}$ for some $r>1$, then $\sum\varepsilon_n$ will converge, and hence $h_\varepsilon\colon n\mapsto n^r$ grows fast enough. We will

see later that the role of the function h is rather unrestrictive in the higher analogue, since, in fact, any choice of h that is eventually nonzero will result in the same family of slaloms (see Lemma 2.5).

We propose two alternative characterisations of $\mathcal{E}$. Let us call a sequence $B = \langle B_n \mid n \in \omega \rangle$ a *bounded partition* of ω, if the sets B_n are mutually disjoint finite sets such that $\omega = \bigcup_{n\in\omega} B_n$. We define $\Sigma(B, \varepsilon)$, $[\varphi]_*$ for $\varphi \in \Sigma(B, \varepsilon)$, and $\mathcal{E}(B, \varepsilon)$ exactly as before, replacing the role of the sets I_n by B_n. This allows us to define $\mathcal{BE} = \bigcup_{B\in \mathrm{BP},\, \varepsilon\in\ell^1_+} \mathcal{E}(B, \varepsilon)$, where BP is the family of bounded partitions of ω.

Theorem 2.2. $\mathcal{E} = \mathcal{BE}$.

Proof. Clearly every interval partition of ω is a bounded partition as well, hence $\mathcal{E} \subseteq \mathcal{BE}$. Reversely, it is easy to see that for every $B \in \mathrm{BP}$, $\varepsilon \in \ell^1_+$ and $\varphi \in \Sigma(B, \varepsilon)$ we have $[\varphi]_* = \bigcup_{m\in\omega}[\varphi]_m$, where we define $[\varphi]_m = \{x \in {}^\omega 2 \mid \forall n \geq m(x \restriction B_n \in \varphi(n))\}$. Now, as each $[\varphi]_m$ is a closed set of measure zero, we have $\mathcal{BE} \subseteq \mathcal{E}$. □

Alternatively, we may consider the following tree-based characterisation of $\mathcal{E}$, which also avoids mention of Lebesgue measure. Let $T \subseteq {}^{<\omega}2$ be a tree and let $[T] \subseteq {}^\omega 2$ denote the set of branches through T. We say that T is a *halving tree* if for every $s \in T$ there is $n_s \in \omega$ such that

$$\Big|\big\{t \in {}^{\mathrm{dom}(s)+n_s}2 \;\big|\; s \subseteq t \in T\big\}\Big| \leq \Big|\big\{t \in {}^{\mathrm{dom}(s)+n_s}2 \;\big|\; s \subseteq t \notin T\big\}\Big|.$$

That is, for each $s \in T$, there is a level of the tree in which at most half of the extensions of s at that level belong to the tree. Let us define $\mathcal{H}$ as the least σ-ideal containing $[T]$ for every halving tree $T \subseteq {}^{<\omega}2$.

Theorem 2.3. $\mathcal{E} = \mathcal{H}$.

Proof. If T is a halving tree and $k \in \omega$, we build a sequence $\langle m_i \in \omega \mid i \leq k\rangle$ with $m_0 = 0$, such that for each $s \in T \cap {}^{m_i}2$ we have $\mathrm{dom}(s) + n_s \leq m_{i+1}$, where n_s is defined as in the halving property. It then follows that $2^{m_k} \geq 2^k \cdot |T \cap {}^{m_k}2|$, thus $\mu(T) \leq 2^{-k}$. This shows that $[T] \in \mathcal{E}$.

On the other hand, if $X \subseteq {}^\omega 2$ is a closed null set, then there is a tree $T \subseteq {}^{<\omega}2$ with $X = [T]$. Let $s \in T$, then there is an open set $U \subseteq [s]$ such that $\mu(U) > \frac{1}{2}\mu([s])$ and $U \cap [T] = \varnothing$. Then we may find $n_s \in \omega$ such that $\mu[s] \leq 2 \cdot \mu(\bigcup\{t \in {}^{\mathrm{dom}(s)+n_s}2 \mid [t] \subseteq U \text{ and } s \subseteq t\})$, from which it follows that T is halving. □

We will see that neither Theorem 2.2 nor Theorem 2.3 applies to the higher context.

2.2. **The higher closed null ideals.** An *interval partition* of κ is a sequence $I = \langle I_\alpha \mid \alpha \in \kappa\rangle$ of mutually disjoint nonempty intervals of κ such that $\kappa = \bigcup_{\alpha\in\kappa} I_\alpha$. Once again we adopt the convention that $\min(I_\alpha) = \min(\kappa \setminus \bigcup_{\xi\in\alpha} I_\xi)$ for each $\alpha \in \kappa$. It is good to note that $\{\min(I_\alpha) \mid \alpha \in \kappa\}$ is a club subset of κ, and equivalently that the function $f\colon \alpha \mapsto \min(I_\alpha)$ is a continuous strictly increasing function. We let IP_κ be the set of all interval partitions of κ. A *bounded partition* of κ is a sequence $B = \langle B_\alpha \mid \alpha \in \kappa\rangle$ of mutually disjoint nonempty bounded subset of κ such that $\kappa = \bigcup_{\alpha\in\kappa} B_\alpha$. We let BP_κ denote the set of all bounded partitions of κ. Note that $\mathrm{IP}_\kappa \subseteq \mathrm{BP}_\kappa$. We will fix the notation that if $I \in \mathrm{IP}_\kappa$ or $B \in \mathrm{BP}_\kappa$, then I_α and B_α are the α-th element of I and B, respectively. Moreover, we write $i_\alpha = \min(I_\alpha)$, whence $I_\alpha = [i_\alpha, i_{\alpha+1})$ for each $\alpha \in \kappa$.

Often, it will be useful to consider only partitions in which the parts become as large as possible. Recall that $\lim_{\alpha<\kappa} \lambda_\alpha = \lambda$ if for every $\mu < \lambda$ we have $\forall^\infty \alpha \in \kappa\, (\mu < \lambda_\alpha)$. Let us define

$$\mathrm{BP}^{\lim}_\kappa = \begin{cases} \{B \in \mathrm{BP}_\kappa \mid \lim_{\alpha<\kappa} |B_\alpha| = \kappa\} & \text{if } \kappa \text{ is a limit cardinal,} \\ \{B \in \mathrm{BP}_\kappa \mid \forall^\infty \alpha \in \kappa\, (|B_\alpha| = \lambda)\} & \text{if } \kappa \text{ is the successor of } \lambda, \end{cases}$$

$$\mathrm{IP}^{\lim}_\kappa = \mathrm{BP}^{\lim}_\kappa \cap \mathrm{IP}_\kappa.$$

Given $h \in {}^{\kappa}(\kappa \setminus \{0\})$ and $B = \langle B_\alpha \mid \alpha \in \kappa \rangle \in \mathrm{BP}_\kappa$, we generalise Equation $(*)$ as:

$$\Sigma^*_\kappa(B,h) = \Big\{ \varphi \in \textstyle\prod_{\alpha\in\kappa} \mathcal{P}\big({}^{B_\alpha}2\big) \;\Big|\; \forall \lambda < \kappa \forall^\infty \alpha \in \kappa \, \big(\lambda \cdot |h(\alpha)| \cdot |\varphi(\alpha)| < 2^{|B_\alpha|}\big) \Big\}.$$

We analogously define, for $\varphi \in \Sigma^*_\kappa(B,h)$ and $\beta \in \kappa$, the families

$$[\varphi]_\beta = \{x \in {}^{\kappa}2 \mid \forall \alpha \in \kappa \setminus \beta \, (x \restriction B_\alpha \in \varphi(\alpha))\},$$
$$[\varphi]_* = \textstyle\bigcup_{\beta\in\kappa} [\varphi]_\beta = \{x \in {}^{\kappa}2 \mid \forall^\infty \alpha \in \kappa \, (x \restriction B_\alpha \in \varphi(\alpha))\}.$$

It is good, at this point, to reflect on our definition for a moment and show that it can be simplified. Specifically, neither λ nor h is relevant for the definition of $\Sigma^*_\kappa(B,h)$, as long as we assume that $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$.

Definition 2.4. For $B \in \mathrm{BP}_\kappa$ let us define

$$\Sigma^*_\kappa(B) = \Big\{ \varphi \in \textstyle\prod_{\alpha\in\kappa} \mathcal{P}\big({}^{B_\alpha}2\big) \;\Big|\; \forall^\infty \alpha \in \kappa \, \big(|\varphi(\alpha)| < 2^{|B_\alpha|}\big) \Big\},$$
$$\Sigma_\kappa(B) = \Big\{ \varphi \in \textstyle\prod_{\alpha\in\kappa} \mathcal{P}\big({}^{B_\alpha}2\big) \;\Big|\; \forall \alpha \in \kappa \, \big(|\varphi(\alpha)| < 2^{|B_\alpha|}\big) \Big\}.$$

Lemma 2.5. Let $h \in {}^{\kappa}(\kappa \setminus \{0\})$ and let $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$, then the following are equivalent:

(1) $\exists \varphi \in \Sigma^*_\kappa(B,h) \, ([\varphi]_* \neq \varnothing)$,
(2) $\forall^\infty \alpha \in \kappa \, (|h(\alpha)| < 2^{|B_\alpha|})$,
(3) $\Sigma^*_\kappa(B,h) = \Sigma^*_\kappa(B)$.

Proof. (1) $\Rightarrow$ (2). Let $\varphi \in \Sigma^*_\kappa(B,h)$ have $[\varphi]_* \neq \varnothing$, then there is $x \in {}^{\kappa}2$ and $\beta \in \kappa$ with $x \in [\varphi]_\beta$. Particularly, $0 < |h(\alpha)| \cdot |\varphi(\alpha)| < 2^{|B_\alpha|}$ for almost all $\alpha \in \kappa$, thus $|h(\alpha)| < 2^{|B_\alpha|}$ for almost all $\alpha \in \kappa$.

(2) $\Rightarrow$ (3). Let $\varphi \in \Sigma^*_\kappa(B,h)$ and let $\lambda = 1$, then $|\varphi(\alpha)| \leq |h(\alpha)| \cdot |\varphi(\alpha)|$ for all $\alpha \in \kappa$, hence $\varphi \in \Sigma^*_\kappa(B)$. On the other hand, since $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$, for any $\lambda < \kappa$ we have $\forall^\infty \alpha \in \kappa \, (|B_\alpha| \geq \lambda)$, hence $\forall^\infty \alpha \in \kappa \, (\lambda < 2^{|B_\alpha|})$, and therefore (2) implies that $\Sigma^*_\kappa(B) \subseteq \Sigma^*_\kappa(B,h)$.

(3) $\Rightarrow$ (1). Since $\lim_{\alpha\in\kappa} |B_\alpha| = \kappa$ implies that there exist $\beta \in \kappa$ and $\varphi \in \Sigma^*_\kappa(B)$ such that $\forall \alpha \geq \beta (\varphi(\alpha) \neq \varnothing)$, it follows that $\varnothing \neq [\varphi]_\beta \subseteq [\varphi]_*$. By (3) we see that $\varphi \in \Sigma^*_\kappa(B,h)$. □

Lemma 2.6. Let $B \in \mathrm{BP}_\kappa$ and $\varphi \in \Sigma^*_\kappa(B)$, then there exists $\varphi' \in \Sigma_\kappa(B)$ such that $[\varphi]_* = [\varphi']_*$.

Proof. It is easy to see that the following φ' suffices:

$$\varphi' \colon \alpha \mapsto \begin{cases} \varphi(\alpha) & \text{if } |\varphi(\alpha)| < 2^{|B_\alpha|}, \\ \varnothing & \text{otherwise.} \end{cases}$$

□

In light of Lemmas 2.5 and 2.6, we will simply work with $\Sigma_\kappa(B)$ from now on.

Definition 2.7. Let $B \in \mathrm{BP}_\kappa$, then we define the following families:

$$\mathcal{E}_\kappa(B) = \{X \subseteq {}^{\kappa}2 \mid \exists \varphi \in \Sigma_\kappa(B)(X \subseteq [\varphi]_*)\},$$
$$\mathcal{E}^\exists_\kappa = \textstyle\bigcup_{I\in \mathrm{IP}^{\mathrm{lim}}_\kappa} \mathcal{E}_\kappa(I), \qquad \mathcal{BE}^\exists_\kappa = \textstyle\bigcup_{B\in \mathrm{BP}^{\mathrm{lim}}_\kappa} \mathcal{E}_\kappa(B),$$
$$\mathcal{E}^\forall_\kappa = \textstyle\bigcap_{I\in \mathrm{IP}^{\mathrm{lim}}_\kappa} \mathcal{E}_\kappa(I), \qquad \mathcal{BE}^\forall_\kappa = \textstyle\bigcap_{B\in \mathrm{BP}^{\mathrm{lim}}_\kappa} \mathcal{E}_\kappa(B),$$
$$\mathcal{E}_\kappa = \Big\{\textstyle\bigcup E \;\Big|\; E \in [\mathcal{E}^\exists_\kappa]^{\leq\kappa}\Big\}, \qquad \mathcal{BE}_\kappa = \Big\{\textstyle\bigcup E \;\Big|\; E \in [\mathcal{BE}^\exists_\kappa]^{\leq\kappa}\Big\}.$$

The following properties are easily verified from the above definition.

Proposition 2.8. The following hold for all $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ and $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$.

(1) $\mathcal{E}^\forall_\kappa \subseteq \mathcal{E}_\kappa(I) \subseteq \mathcal{E}^\exists_\kappa \subseteq \mathcal{E}_\kappa$,
(2) $\mathcal{BE}^\forall_\kappa \subseteq \mathcal{E}_\kappa(B) \subseteq \mathcal{BE}^\exists_\kappa \subseteq \mathcal{BE}_\kappa$,
(3) $\mathcal{BE}^\forall_\kappa \subseteq \mathcal{E}^\forall_\kappa$ and $\mathcal{E}^\exists_\kappa \subseteq \mathcal{BE}^\exists_\kappa$ and $\mathcal{E}_\kappa \subseteq \mathcal{BE}_\kappa$.

(4) $\mathcal{E}_\kappa$ and $\mathcal{BE}_\kappa$ are $\leq\kappa$-complete ideals. □

We will show in Proposition 2.9 that when $B \in \mathrm{BP}_\kappa^{\mathrm{lim}}$ also $\mathcal{E}_\kappa(B)$, $\mathcal{E}_\kappa^\forall$ and $\mathcal{BE}_\kappa^\forall$ are $\leq\kappa$-complete ideals. In Section 2.4, we study the relations between these $\mathcal{E}_\kappa(B)$ for different partitions $B \in \mathrm{BP}_\kappa$; particularly we will see in Proposition 2.24 that no $\mathcal{E}_\kappa(B)$ is $\subseteq$-least with respect to all ideals of the form $\mathcal{E}_\kappa(B)$, which implies that $\mathcal{E}_\kappa^\forall \neq \mathcal{E}_\kappa(I)$ and $\mathcal{BE}_\kappa^\forall \neq \mathcal{E}_\kappa(B)$ for any $I \in \mathrm{IP}_\kappa^{\mathrm{lim}}$ and $B \in \mathrm{BP}_\kappa^{\mathrm{lim}}$. We also mention that not all of the families from Definition 2.7 are ideals, as we will see in Theorem 2.28 that $\mathcal{BE}_\kappa^\exists$ is not closed under finite unions. For strongly inaccessible κ, also $\mathcal{E}_\kappa^\exists$ is not an ideal, as shown in [unresolved original author label: thm:Ekm not ideal], but $\mathcal{E}_\kappa^\exists = \mathcal{E}_\kappa$ if κ is successor by Corollary 2.25. Consequently, all inclusions in (1) and (2) are strict for certain κ. Finally, the last two inclusions in (3) will be shown to be strict as well, in Theorem 2.10. The first inclusion in (3) is an equality if κ is successor, as we will see in Theorem 2.33. See also Figure 1.

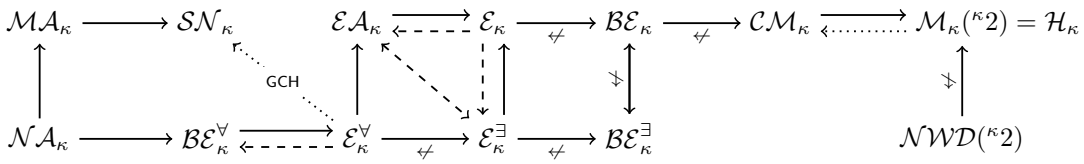


FIGURE 1. Comparison of families of subsets of ${}^\kappa 2$, where an arrow indicates inclusion. An inclusion is proper if the corresponding arrow is labelled with a small negated arrow in the other direction. Dotted arrows are provable if κ is inaccessible, dashed arrows are provable if κ is successor. For definitions, see Definitions 2.7, 2.11 and 2.17 and Section 3.1.

Proposition 2.9.

(1) If $B \in \mathrm{BP}_\kappa$ and $\lambda := \liminf_{\alpha\in\kappa} |B_\alpha|$ is infinite, then $\mathcal{E}_\kappa(B)$ is a $\leq\lambda$-complete ideal.
(2) If $B \in \mathrm{BP}_\kappa^{\mathrm{lim}}$, then $\mathcal{E}_\kappa(B)$ is a $\leq\kappa$-complete ideal.
(3) $\mathcal{E}_\kappa^\forall$ and $\mathcal{BE}_\kappa^\forall$ are $\leq\kappa$-complete ideals.

Proof. (1). Let $B \in \mathrm{BP}_\kappa$, let $\langle \varphi_\xi \mid \xi \in \lambda\rangle$ be a sequence of slaloms in $\Sigma_\kappa(B)$, and define

$$\varphi\colon \alpha \mapsto \bigcup\{\varphi_\xi(\alpha) \mid \xi < \min(\lambda, |B_\alpha|) \text{ and } |\varphi_\xi(\alpha)| < 2^{|B_\alpha|}\}.$$

If $|B_\alpha|$ is infinite, then $|B_\alpha| < \mathrm{cf}(2^{|B_\alpha|})$ implies that $|\varphi(\alpha)| < 2^{|B_\alpha|}$. But this applies to almost all $\alpha \in \kappa$, since $\liminf_{\alpha\in\kappa}|B_\alpha| = \lambda$ is infinite and κ is regular. Moreover, for almost all $\alpha \in \kappa$ we see that $\varphi_\xi(\alpha) \subseteq \varphi(\alpha)$, and thus it follows that $\bigcup_{\xi\in\lambda}[\varphi_\xi]_* \subseteq [\varphi]_*$.

(2). If κ is regular limit, this follows from (1). Suppose $\kappa = \lambda^+$ and let $B \in \mathrm{BP}_\kappa^{\mathrm{lim}}$, then we may assume that $|B_\alpha| = \lambda$ for all $\alpha \in \kappa$. Let $\varphi_\xi \in \Sigma_\kappa(B)$ for each $\xi \in \kappa$, and note that $\kappa = 2^{<\kappa} = 2^\lambda$ implies $|\varphi_\xi(\alpha)| \leq \lambda$ for each $\alpha \in \kappa$. Then, particularly, if we define $\varphi : \alpha \mapsto \bigcup_{\xi\in\alpha} \varphi_\xi(\alpha)$, we see that $|\varphi(\alpha)| \leq \lambda$ for each $\alpha \in \kappa$, and thus $\varphi \in \Sigma_\kappa(B)$, and $\bigcup_{\xi\in\kappa}[\varphi_\xi]_* \subseteq [\varphi]_*$.

(3). This is clear, since both $\mathcal{BE}_\kappa^\forall$ and $\mathcal{E}_\kappa^\forall$ are the intersection of a family of $\leq\kappa$-complete ideals. □

Theorem 2.10. Let κ be regular uncountable, then $\mathcal{BE}_\kappa^\exists \not\subseteq \mathcal{E}_\kappa$. Hence, $\mathcal{E}_\kappa^\exists \subsetneq \mathcal{BE}_\kappa^\exists$ and $\mathcal{E}_\kappa \subsetneq \mathcal{BE}_\kappa$.

Proof. The inclusions easily follow from $\mathrm{IP}_\kappa^{\mathrm{lim}} \subseteq \mathrm{BP}_\kappa^{\mathrm{lim}}$.

Let $E = \{2\cdot\alpha \mid \alpha \in \kappa\}$ and $X = \{x \in {}^\kappa 2 \mid x^{-1}(1) \subseteq E\}$. That is, E is the set of even ordinals below κ, and X is the set of $x \in {}^\kappa 2$ that map all odd ordinals to 0. We claim that $X \in \mathcal{BE}_\kappa^\exists \setminus \mathcal{E}_\kappa$.

On the one hand, we may choose $B \in \mathrm{BP}^{\lim}_\kappa$ such that for each $\alpha \in \kappa$ there exists ξ_α with $E \cap B_\alpha = \{\xi_\alpha\}$, i.e., each B_α contains exactly one even ordinal. Let us define

$$\varphi\colon \alpha \mapsto \big\{ s \in {}^{B_\alpha}2 \;\big|\; \forall \xi \in B_\alpha\, (\xi \neq \xi_\alpha \to s(\xi) = 0)\big\},$$

then $|\varphi(\alpha)| = 2$ and $X \subseteq [\varphi]_* \in \mathcal{E}_\kappa(B) \subseteq \mathcal{BE}^\exists_\kappa$.

On the other hand, suppose $Y \in \mathcal{E}_\kappa$, then there are $I^\zeta \in \mathrm{IP}^{\lim}_\kappa$ and $\varphi^\zeta \in \Sigma_\kappa(I^\zeta)$ for each $\zeta \in \kappa$ such that $Y \subseteq \bigcup_{\zeta\in\kappa}[\varphi^\zeta]_*$. For each $\alpha \in \kappa$ choose β_α, ζ_α such that

(1) the intervals $I_\alpha := I^{\zeta_\alpha}_{\beta_\alpha}$ are infinite and pairwise disjoint,
(2) $\big|\{\alpha \in \kappa \mid \zeta = \zeta_\alpha\}\big| = \kappa$ for every $\zeta \in \kappa$.

Since I_α is an infinite interval, $|X \restriction I_\alpha| = 2^{|I_\alpha|}$, thus we may choose some $s_\alpha \in (X \restriction I_\alpha) \setminus \varphi^{\zeta_\alpha}(\beta_\alpha)$. Now note that there exists $x \in X$ that is an extension of $\bigcup_{\alpha\in\kappa} s_\alpha$ and clearly $x \in X \setminus Y$. □

We will show that reverse inclusion does not hold for strongly inaccessible κ in [unresolved original author label: thm:Ekm not ideal].

2.3. **Relations with the higher meagre ideal.** We will now compare our ideals $\mathcal{E}_\kappa$ and $\mathcal{BE}_\kappa$ to the κ-meagre ideal and show that $\mathcal{BE}_\kappa \subsetneq \mathcal{M}_\kappa({}^\kappa 2)$. In fact, we will show a slightly stronger result. For this, we will introduce the ideal of *combinatorially κ-meagre* sets, due to Blass et al. [7].

Definition 2.11. Let $x \in {}^\kappa 2$ and $B \in \mathrm{BP}_\kappa$, then we define the sets

$$\langle\!\langle x, B\rangle\!\rangle = \{y \in {}^\kappa 2 \mid \forall \alpha \in \kappa\, (y \restriction B_\alpha \neq x \restriction B_\alpha)\},$$
$$\langle\!\langle x, B\rangle\!\rangle_* = \{y \in {}^\kappa 2 \mid \forall^\infty \alpha \in \kappa\, (y \restriction B_\alpha \neq x \restriction B_\alpha)\}.$$

A set $Y \subseteq {}^\kappa 2$ is *combinatorially κ-meagre* if there exist $x \in {}^\kappa 2$ and $I \in \mathrm{IP}_\kappa$ such that $Y \subseteq \langle\!\langle x, I\rangle\!\rangle_*$. Let $\mathcal{CM}_\kappa$ denote the family of combinatorially κ-meagre sets.

Proposition 2.12 (Blass et al. [7, § 4])**.** Let κ be regular uncountable, then

(1) $\mathcal{CM}_\kappa$ is a $\leq\kappa$-complete ideal,
(2) $\mathcal{CM}_\kappa \subseteq \mathcal{M}_\kappa({}^\kappa 2)$,
(3) $\mathcal{CM}_\kappa = \mathcal{M}_\kappa({}^\kappa 2)$ if and only if κ is strongly inaccessible.

Note that we defined combinatorially κ-meagre sets only using interval partitions, but that we may as well have used bounded partitions instead, as a corollary of the following observation.

Lemma 2.13. If $B \in \mathrm{BP}_\kappa$, then there exist $I \in \mathrm{IP}_\kappa$ and a sequence of sets $\langle A_\alpha \mid \alpha \in \kappa\rangle$ such that $I_\alpha = \bigcup_{\beta\in A_\alpha} B_\beta$ for each $\alpha \in \kappa$.

Proof. One constructs I_α, A_α simultaneously by recursion. Given that we have constructed I_ξ, A_ξ for each $\xi < \alpha$, let $\gamma_0 = \bigcup_{\xi<\alpha} I_\xi$ (which is an ordinal) and let $A^0_\alpha = \{\beta\}$ where β is such that $\gamma_0 \in B_\beta$. For each $n \in \omega$, let γ_{n+1} be the least ordinal such that $\forall \gamma \in \bigcup_{\beta\in A^n_\alpha} B_\beta\, (\gamma_{n+1} > \gamma)$, and define

$$A^{n+1}_\alpha = \{\beta \in \kappa \mid \exists \gamma \in [\gamma_0, \gamma_{n+1})(\gamma \in B_\beta)\}.$$

Finally, $A_\alpha = \bigcup_{n\in\omega} A^n_\alpha$ and $I_\alpha = [\gamma_0, \bigcup_{n\in\omega} \gamma_n)$. □

Corollary 2.14. $\mathcal{CM}_\kappa = \{Y \subseteq {}^\kappa 2 \mid \exists x \in {}^\kappa 2\, \exists B \in \mathrm{BP}_\kappa\, (Y \subseteq \langle\!\langle x, B\rangle\!\rangle_*)\}$.

In order to compare $\mathcal{BE}_\kappa$ to $\mathcal{CM}_\kappa$, we will be considering $\langle\!\langle x, I\rangle\!\rangle$ for $x \in {}^\kappa 2$ and $I \in \mathrm{IP}_\kappa$ as a subspace of the higher Cantor space ${}^\kappa 2$. Particularly, we will use the κ-meagre ideal $\mathcal{M}_\kappa(\langle\!\langle x, I\rangle\!\rangle)$ of the space $\langle\!\langle x, I\rangle\!\rangle$. First we note that the κ-Baire category theorem holds in $\langle\!\langle x, I\rangle\!\rangle$:

Lemma 2.15. Let $x \in {}^{\kappa}2$ and let $I \in \mathrm{IP}_\kappa$ be such that $|I_\alpha| \geq 2$ for all $\alpha \in \kappa$, then $\mathcal{M}_\kappa(\langle\!\langle x, I\rangle\!\rangle)$ is a proper ideal on $\langle\!\langle x, I\rangle\!\rangle$.

Proof. Let $X_\alpha = {}^{I_\alpha}2 \setminus \{x \restriction I_\alpha\}$ for $\alpha \in \kappa$ and consider the higher product space $\mathcal{X} = \prod_{\alpha\in\kappa} X_\alpha$. Since $|X_\alpha| \geq 2$ for all $\alpha \in \kappa$, one easily sees that $\mathcal{X} \notin \mathcal{M}_\kappa(\mathcal{X})$, as in the classical proof of the Baire category theorem. Finally note that $\langle\!\langle x, I\rangle\!\rangle$ is homeomorphic to $\mathcal{X}$. □

Theorem 2.16. Let κ be regular uncountable, then $\mathcal{BE}_\kappa \subsetneq \mathcal{CM}_\kappa$.

Proof. Let $Y \in \mathcal{BE}_\kappa^{\exists}$, let $B \in \mathrm{BP}_\kappa^{\mathrm{lim}}$ and let $\varphi \in \Sigma_\kappa(B)$ be such that $Y \subseteq [\varphi]_*$. For each $\alpha \in \kappa$ pick $s_\alpha \in {}^{B_\alpha}2$ such that $s_\alpha \notin \varphi(\alpha)$ for almost all $\alpha \in \kappa$, and define $x = \bigcup_{\alpha\in\kappa} s_\alpha$. If $y \in Y$, then $y \restriction B_\alpha \in \varphi(\alpha)$ for almost all α, hence $y \restriction B_\alpha \neq x \restriction B_\alpha$ for almost all α, thus $y \in \langle\!\langle x, B\rangle\!\rangle_*$. Therefore $Y \in \mathcal{CM}_\kappa$, and thus $\mathcal{BE}_\kappa^{\exists} \subseteq \mathcal{CM}_\kappa$. By Proposition 2.12 (1), it follows that $\mathcal{BE}_\kappa \subseteq \mathcal{CM}_\kappa$.

Let $x \in {}^{\kappa}2$ and $I \in \mathrm{IP}_\kappa^{\mathrm{lim}}$ with $|I_\alpha| \geq 2$ for all $\alpha \in \kappa$, then $\langle\!\langle x, I\rangle\!\rangle \in \mathcal{CM}_\kappa \setminus \mathcal{M}_\kappa(\langle\!\langle x, I\rangle\!\rangle)$ by Lemma 2.15 and the definition of $\mathcal{CM}_\kappa$. To conclude the proof, it suffices to show that $A \cap \langle\!\langle x, I\rangle\!\rangle \in \mathcal{M}_\kappa(\langle\!\langle x, I\rangle\!\rangle)$ for all $A \in \mathcal{BE}_\kappa^{\exists}$. This also implies for all $A' \in \mathcal{BE}_\kappa$ that $A' \cap \langle\!\langle x, I\rangle\!\rangle \in \mathcal{M}_\kappa(\langle\!\langle x, I\rangle\!\rangle)$, since $\mathcal{M}_\kappa(\langle\!\langle x, I\rangle\!\rangle)$ is $\leq\kappa$-complete. Therefore $\langle\!\langle x, I\rangle\!\rangle \notin \mathcal{BE}_\kappa$. Fix some $B \in \mathrm{BP}_\kappa^{\mathrm{lim}}$ and $\varphi \in \Sigma_\kappa(B)$, then we show that $[\varphi]_\beta \cap \langle\!\langle x, I\rangle\!\rangle \in \mathcal{NWD}(\langle\!\langle x, I\rangle\!\rangle)$ for each $\beta < \kappa$, from which it follows that $[\varphi]_* \cap \langle\!\langle x, I\rangle\!\rangle \in \mathcal{M}_\kappa(\langle\!\langle x, I\rangle\!\rangle)$.

Fix some $s \in {}^{<\kappa}2$ such that $\mathrm{dom}(s) = \bigcup_{\alpha\in\delta} I_\alpha$ for some $\delta \in \kappa$ and $[s] \cap \langle\!\langle x, I\rangle\!\rangle \neq \varnothing$. Let $\gamma \geq \beta$ and $\delta' > \delta$ be such that B_γ is infinite and $B_\gamma \subseteq \bigcup_{\alpha\in[\delta,\,\delta')} I_\alpha$. For each $\alpha \in [\delta, \delta')$, pick $\xi_\alpha \in I_\alpha$ with the only restriction that $\xi_\alpha \notin B_\gamma$ whenever $I_\alpha \nsubseteq B_\gamma$. Now note that $|B_\gamma \setminus \{\xi_\alpha \mid \alpha \in [\delta, \delta')\}| = |B_\gamma|$, so there is $t \in {}^{B_\gamma}2 \setminus \varphi(\gamma)$ such that $t(\xi_\alpha) \neq x(\xi_\alpha)$ for all $\alpha \in [\delta, \delta')$. We may now extend $s \cup t$ to some r with $\mathrm{dom}(r) = \bigcup_{\alpha\in\delta'} I_\alpha$ where $r(\xi_\alpha) \neq x(\xi_\alpha)$ for all $\alpha \in [\delta, \delta')$. Then $[r] \cap \langle\!\langle x, I\rangle\!\rangle \neq \varnothing$ and also $[r] \cap [\varphi]_\beta = \varnothing$. □

As a consequence of Theorem 2.16, we observe that $\mathcal{NWD}({}^{\kappa}2) \nsubseteq \mathcal{BE}_\kappa$, since the set $\langle\!\langle x, I\rangle\!\rangle$ from the above proof is nowhere dense in ${}^{\kappa}2$. Reversely, we can also observe that $\mathcal{BE}_\kappa^{\forall} \nsubseteq \mathcal{NWD}({}^{\kappa}2)$: because $\kappa = 2^{<\kappa}$, it follows that $[{}^{\kappa}2]^\kappa$ contains a dense set, and we have $[{}^{\kappa}2]^\kappa \subseteq \mathcal{BE}_\kappa^{\forall}$.

Now let us take a brief look at the generalisation of halving trees and the ideal $\mathcal{H}$ from Theorem 2.3.

Definition 2.17. Call a tree $T \subseteq {}^{<\kappa}2$ a *halving tree* if for every $s \in T$ there is $\alpha_s \in \kappa$ such that

$$\Big|\big\{t \in {}^{\mathrm{dom}(s)+\alpha_s}2 \;\big|\; s \subseteq t \in T\big\}\Big| \leq \Big|\big\{t \in {}^{\mathrm{dom}(s)+\alpha_s}2 \;\big|\; s \subseteq t \notin T\big\}\Big|.$$

Let $\mathcal{H}_\kappa$ be the least κ-complete ideal containing $[T]$ for every halving tree $T \subseteq {}^{<\kappa}2$.

In contrast with the classical context, this ideal is actually equal to the κ-meagre ideal:

Theorem 2.18. $\mathcal{M}_\kappa({}^\kappa 2) = \mathcal{H}_\kappa$.

Proof. Easily $[T] \in \mathcal{NWD}({}^\kappa 2)$ for every halving $T \subseteq {}^{<\kappa}2$. We show that reversely $\mathcal{NWD}({}^\kappa 2) \subseteq \mathcal{H}_\kappa$.

Let $N \subseteq {}^\kappa 2$ be nowhere dense, and without loss of generality, we may assume N is closed. Let $T \subseteq {}^{<\kappa}2$ be such that $N = [T]$ and $s \in T$, then there is $t \in {}^{<\kappa}2$ with $s \subseteq t$ and $t \notin T$. We may also assume without loss of generality that $\mathrm{dom}(t) \geq \mathrm{dom}(s) + \gamma$ for some infinite γ, then $\alpha_s = \gamma \cdot 2$ suffices: there are $|{}^\gamma 2| = 2^{|\gamma|}$ extensions of t at the level $\mathrm{dom}(s) + \alpha_s$, but s also has $|{}^{\gamma\cdot 2}2| = 2^{|\gamma|}$ extensions at this level. □

2.4. **Comparing the generating ideals.** Remember the classical Theorem 2.1:

$$\mathcal{E} = \bigcup_{I \in \mathrm{IP}, \varepsilon \in \ell^1_+} \mathcal{E}(I, \varepsilon).$$

Since $\mathcal{E}$ is a σ-ideal by definition, it follows from this theorem that, given $I^n \in \mathrm{IP}$ and $\varepsilon^n \in \ell^1_+$ and slaloms $\varphi^n \in \Sigma(I^n, \varepsilon^n)$ for each $n \in \omega$, there are $I \in \mathrm{IP}$ and $\varepsilon \in \ell^1_+$ such that $\bigcup_{n\in\omega}[\varphi^n]_* \in \mathcal{E}(I, \varepsilon)$. Exactly, the theorem implies that any countably many slaloms are contained in a single generating set $\mathcal{E}(I, \varepsilon)$ of the ideal $\mathcal{E}$.

In this section we show that the higher analogue to Theorem 2.1 is false. In fact, we will show that $\mathcal{BE}^\exists_\kappa$ and $\mathcal{E}^\exists_\kappa$ are not even ideals, as they are not closed under finite unions. We also present various results comparing the generating sets $\mathcal{E}_\kappa(B)$ for different $B \in \mathrm{BP}^{\lim}_\kappa$.

Consider the equivalence relation $=^*$ on ${}^\kappa 2$, where $f =^* g$ if $\forall^\infty \alpha \in \kappa\,(f(\alpha) = g(\alpha))$. If $\varphi \in \Sigma_\kappa(B)$ for some $B \in \mathrm{BP}_\kappa$, then $[\varphi]_*$ is closed under $=^*$, in the sense that $f \in [\varphi]_*$ and $g \in {}^\kappa 2$ with $f =^* g$ implies that $g \in [\varphi]_*$ as well. For $A \subseteq {}^\kappa 2$, let us define $(A)_* = \{f \in {}^\kappa 2 \mid \exists g \in A\,(f =^* g)\}$. We observe the following basic properties.

Proposition 2.19. Let $A, A' \subseteq {}^\kappa 2$, $B \in \mathrm{BP}_\kappa$, $\varphi \in \Sigma_\kappa(B)$, and $\beta \in \kappa$, then

(1) $A \subseteq (A)_*$ and if $A \subseteq A'$, then $(A)_* \subseteq (A')_*$,
(2) $([\varphi]_\beta)_* = ([\varphi]_*)_* = [\varphi]_*$,
(3) $A \in \mathcal{E}_\kappa(B)$ if and only if $(A)_* \in \mathcal{E}_\kappa(B)$,
(4) $A \subseteq [\varphi]_*$ if and only if $(A)_* \subseteq [\varphi]_*$. □

For a fixed $B \in \mathrm{BP}_\kappa$ and $\varphi \in \Sigma_\kappa(B)$, we may consider $[\varphi]_0 = \{x \in {}^\kappa 2 \mid \forall \alpha \in \kappa\,(x \restriction B_\alpha \in \varphi(\alpha))\}$ as a subspace of ${}^\kappa 2$, and naturally we may define the κ-meagre ideal $\mathcal{M}_\kappa([\varphi]_0)$ on this space, similar to what we did in Lemma 2.15. Note, however, that $\mathcal{M}_\kappa([\varphi]_0)$ need not be proper for certain pathological choices of φ (and B). E.g., if $\varphi(\alpha)$ is a singleton for all $\alpha \in \kappa$, then $[\varphi]_0$ is the one-point space.

Lemma 2.20. Let $B \in \mathrm{BP}_\kappa$ and $\varphi \in \Sigma_\kappa(B)$ be such that $\exists^\infty \alpha \in \kappa\,(|\varphi(\alpha)| > 1)$, then $\mathcal{M}_\kappa([\varphi]_0)$ is a proper $\leq\kappa$-complete ideal on $[\varphi]_0$.

Proof. This follows quickly from the fact that $[\varphi]_0$ is closed. □

Observe that the conditions of the above lemma can only be fulfilled if $\exists^\infty \alpha \in \kappa\,(|B_\alpha| > 1)$. We are now ready to give a characterisation of the conditions on $B, C \in \mathrm{BP}_\kappa$ for which $\mathcal{E}_\kappa(B) \subseteq \mathcal{E}_\kappa(C)$.

Lemma 2.21. Let $B, C \in \mathrm{BP}_\kappa$ and for $\alpha \in \kappa$ let $S_\alpha = \{\delta \in \kappa \mid B_\delta \cap C_\alpha \neq \varnothing\}$. The following conditions are equivalent:

(1) $\mathcal{E}_\kappa(B) \subseteq \mathcal{E}_\kappa(C)$,
(2) for all $\varphi \in \Sigma_\kappa(B)$ there exists $\varphi' \in \Sigma_\kappa(C)$ such that $[\varphi]_* \subseteq [\varphi']_*$,

(3) for all $\varphi \in \Sigma_\kappa(B)$ for which $\exists^\infty \alpha \in \kappa\,(|\varphi(\alpha)| > 1)$ holds, there exists $\varphi' \in \Sigma_\kappa(C)$ such that $[\varphi']_* \cap [\varphi]_0 \notin \mathcal{M}_\kappa([\varphi]_0)$,

(4) $\forall^\infty \alpha \in \kappa \left(\prod_{\delta\in S_\alpha} \kappa_{\alpha,\delta} < 2^{|C_\alpha|}\right)$ holds for any system $\langle \kappa_{\alpha,\delta} \mid \alpha \in \kappa,\, \delta \in S_\alpha\rangle$ of cardinals such that $\kappa_{\alpha,\delta} < 2^{|B_\delta|}$ and $\kappa_{\alpha,\delta} \leq 2^{|B_\delta \cap C_\alpha|}$ for all $\delta \in S_\alpha$ and $\alpha \in \kappa$.

Proof. (1) $\Leftrightarrow$ (2). Obvious.

(2) $\Rightarrow$ (3). Let $\varphi \in \Sigma_\kappa(B)$ be such that $\exists^\infty \alpha \in \kappa\,(|\varphi(\alpha)| > 1)$ holds. By (2), there exists some $\varphi' \in \Sigma_\kappa(C)$ such that $[\varphi]_0 \subseteq [\varphi]_* \subseteq [\varphi']_*$, and thus $[\varphi']_* \cap [\varphi]_0 = [\varphi]_0$. By Lemma 2.20, we have $[\varphi]_0 \notin \mathcal{M}_\kappa([\varphi]_0)$.

(3) $\Rightarrow$ (4). We prove the contrapositive. That is, we will construct $\varphi \in \Sigma_\kappa(B)$ such that for every $\varphi' \in \Sigma_\kappa(C)$ we have $[\varphi']_* \cap [\varphi]_0 \in \mathcal{M}_\kappa([\varphi]_0)$, under the assumption that there is $K \in [\kappa]^\kappa$ such that $2^{|C_\alpha|} \leq \prod_{\delta\in S_\alpha} \kappa_{\alpha,\delta}$ for all $\alpha \in K$, where $\langle \kappa_{\alpha,\delta} \mid \alpha \in \kappa,\, \delta \in S_\alpha\rangle$ is such that $\kappa_{\alpha,\delta} < 2^{|B_\delta|}$ and $\kappa_{\alpha,\delta} \leq 2^{|B_\delta\cap C_\alpha|}$ for all $\delta \in S_\alpha$ and $\alpha \in \kappa$.

We may assume, by shrinking K if necessary, that $S_\alpha \cap S_{\alpha'} = \varnothing$ for all distinct $\alpha, \alpha' \in K$. For each $\alpha \in K$ and $\delta \in S_\alpha$, we have $\kappa_{\alpha,\delta} \leq 2^{|B_\delta\cap C_\alpha|}$, thus we may choose $F_\delta \subseteq {}^{B_\delta}2$ such that $\kappa_{\alpha,\delta} = |F_\delta| = \big|\{s \restriction (B_\delta \cap C_\alpha) \mid s \in F_\delta\}\big|$. Furthermore, since $\kappa_{\alpha,\delta} < 2^{|B_\delta|}$, we may choose some $\varphi \in \Sigma_\kappa(B)$ such that $\varphi(\delta) = F_\delta$ for all $\delta \in \bigcup_{\alpha\in K} S_\alpha$. Note that for any $\alpha \in K$, we have $2 \leq 2^{|C_\alpha|} \leq \prod_{\delta\in S_\alpha} \kappa_{\alpha,\delta}$, thus there is $\delta \in S_\alpha$ such that $2 \leq \kappa_{\alpha,\delta}$, and therefore Lemma 2.20 tells us that $\mathcal{M}_\kappa([\varphi]_0)$ is a proper ideal on $[\varphi]_0$.

Let $\varphi' \in \Sigma_\kappa(C)$ and $\beta \in \kappa$ be arbitrary, then we show that $[\varphi']_\beta \cap [\varphi]_0 \in \mathcal{NWD}([\varphi]_0)$, and consequently $[\varphi']_* \cap [\varphi]_0 \in \mathcal{M}_\kappa([\varphi]_0)$. Let $s \in {}^{<\kappa}2$ be such that $[s] \cap [\varphi]_0 \neq \varnothing$, then we may choose $\alpha \in K \setminus \beta$ such that $\mathrm{dom}(s) \cap \bigcup_{\delta\in S_\alpha} B_\delta = \varnothing$. By our above construction

$$2^{|C_\alpha|} \leq \prod_{\delta\in S_\alpha} \kappa_{\alpha,\delta} = \prod_{\delta\in S_\alpha} \big|\{s \restriction (B_\delta \cap C_\alpha) \mid s \in \varphi(\delta)\}\big|,$$

hence we may choose $t_\delta \in \varphi(\delta)$ for each $\delta \in S_\alpha$ such that $t = \bigcup_{\delta\in S_\alpha} s_\delta$ has the property that $t \restriction C_\alpha \notin \varphi'(\alpha)$. Finally, there is $x \in [s] \cap [\varphi]_0$ with $t \subseteq x$, and thus if γ is large enough such that $\bigcup_{\delta\in S_\alpha} B_\delta \subseteq \gamma$, then $[x \restriction \gamma] \cap [\varphi]_0 \neq \varnothing$, and because $x \restriction C_\alpha \notin \varphi'(\alpha)$ we also have $[x \restriction \gamma] \cap [\varphi']_\beta = \varnothing$.

(4) $\Rightarrow$ (2). Let $\varphi \in \Sigma_\kappa(B)$ be given and let $F_{\alpha,\delta} = \{s \restriction (B_\delta \cap C_\alpha) \mid s \in \varphi(\delta)\}$ and $\kappa_{\alpha,\delta} = |F_{\alpha,\delta}|$ for each $\alpha \in \kappa$ and $\delta \in S_\alpha$. Define

$$\varphi'(\alpha) = \big\{s \in {}^{C_\alpha}2 \;\big|\; \forall \delta \in S_\alpha\,(s \restriction (B_\delta \cap C_\alpha) \in F_{\alpha,\delta})\big\},$$

then $[\varphi]_* \subseteq [\varphi']_*$ and $|\varphi'(\alpha)| = \prod_{\delta\in S_\alpha} \kappa_{\alpha,\delta}$. Hence, by (4), we see that $|\varphi'(\alpha)| < 2^{|C_\alpha|}$ for almost all $\alpha \in \kappa$, which means that $\varphi' \in \Sigma^*_\kappa(C)$. By Lemma 2.6, we are done. □

Corollary 2.22. Let $B, C \in \mathrm{BP}_\kappa$. Let $S_\alpha = \{\delta \in \kappa \mid B_\delta \cap C_\alpha \neq \varnothing\}$.

(1) If $\forall^\infty \delta \in \kappa\,(|B_\delta| \geq 2)$ and $\exists^\infty \alpha \in \kappa\,(|S_\alpha| = |C_\alpha|)$, then $\mathcal{E}_\kappa(B) \not\subseteq \mathcal{E}_\kappa(C)$.

(2) If $\exists^\infty \alpha \in \kappa\, \forall \delta \in \kappa \left(2^{|B_\delta\cap C_\alpha|} < 2^{|B_\delta|}\right)$, then $\mathcal{E}_\kappa(B) \not\subseteq \mathcal{E}_\kappa(C)$.

Proof. (1). Note that $2 \leq 2^{B_\delta\cap C_\alpha}$ for all $\alpha \in \kappa$ and $\delta \in S_\alpha$, and that by assumption $2 < 2^{|B_\delta|}$ for almost all $\delta \in \kappa$. Therefore, if we let $\kappa_{\alpha,\delta} = 2$ for all $\alpha \in \kappa$ and $\delta \in S_\alpha$, then $\exists^\infty \alpha \in \kappa\,(|S|_\alpha = |C_\alpha|)$ implies that $\exists^\infty \alpha \in \kappa \left(\prod_{\delta\in S_\alpha} \kappa_{\alpha,\delta} = 2^{|S_\alpha|} = 2^{|C_\alpha|}\right)$. By Lemma 2.21, we see that $\mathcal{E}_\kappa(B) \not\subseteq \mathcal{E}_\kappa(C)$.

(2). This follows from Lemma 2.21 by letting $\kappa_{\alpha,\delta} = 2^{|B_\delta\cap C_\alpha|}$. □

Corollary 2.23. If $B, C \in \mathrm{BP}^{\lim}_\kappa$ and there is some regular λ with $\lambda < \kappa$ such that $|B_\delta \cap C_\alpha| < \lambda$ for almost all $\alpha, \delta \in \kappa$, then $\mathcal{E}_\kappa(B) \not\subseteq \mathcal{E}_\kappa(C)$ and $\mathcal{E}_\kappa(C) \not\subseteq \mathcal{E}_\kappa(B)$.

Proof. Since $C \in \mathrm{BP}^{\lim}_\kappa$, we see that $\forall^\infty \alpha \in \kappa\,(|C_\alpha| \geq \lambda)$. Hence, for large enough α, we have $|S_\alpha| \geq \lambda$, where $S_\alpha = \{\delta \in \kappa \mid B_\delta \cap C_\alpha \neq \varnothing\}$. It follows from Corollary 2.22 that $\mathcal{E}_\kappa(B) \not\subseteq \mathcal{E}_\kappa(C)$. The reverse follows by a symmetric argument. □

We will use the above corollary to make some notes about the existence of greatest, least and incomparable elements of the partial orders $\langle\{\mathcal{E}_\kappa(B) \mid B \in \mathrm{BP}^{\mathrm{lim}}_\kappa\}, \subseteq\rangle$ and $\langle\{\mathcal{E}_\kappa(I) \mid I \in \mathrm{IP}^{\mathrm{lim}}_\kappa\}, \subseteq\rangle$.

Proposition 2.24.

(1) If κ is regular uncountable, then $\langle\{\mathcal{E}_\kappa(B) \mid B \in \mathrm{BP}^{\mathrm{lim}}_\kappa\}, \subseteq\rangle$ and $\langle\{\mathcal{E}_\kappa(I) \mid I \in \mathrm{IP}^{\mathrm{lim}}_\kappa\}, \subseteq\rangle$ have no least element.

(2) If κ is regular uncountable, then for every $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$ there exists $C \in \mathrm{BP}^{\mathrm{lim}}_\kappa$ such that $\mathcal{E}_\kappa(B)$ and $\mathcal{E}_\kappa(C)$ are incomparable. Hence, $\langle\{\mathcal{E}_\kappa(B) \mid B \in \mathrm{BP}^{\mathrm{lim}}_\kappa\}, \subseteq\rangle$ has no greatest element.

(3) If κ is regular limit, then for every $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ there exists $J \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ such that $\mathcal{E}_\kappa(I)$ and $\mathcal{E}_\kappa(J)$ are incomparable. Hence, $\langle\{\mathcal{E}_\kappa(I) \mid I \in \mathrm{IP}^{\mathrm{lim}}_\kappa\}, \subseteq\rangle$ has no greatest element.

(4) If κ is successor, then $\langle\{\mathcal{E}_\kappa(I) \mid I \in \mathrm{IP}^{\mathrm{lim}}_\kappa\}, \subseteq\rangle$ has a greatest element.

Proof. (1). We will show this for $\mathrm{BP}^{\mathrm{lim}}_\kappa$, but the situation for $\mathrm{IP}^{\mathrm{lim}}_\kappa$ is identical. Suppose that $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$, then we construct $C \in \mathrm{BP}^{\mathrm{lim}}_\kappa$ such that $\mathcal{E}_\kappa(B) \nsubseteq \mathcal{E}_\kappa(C)$. Note that we can recursively construct a continuous strictly increasing sequence of ordinals $\langle\xi_\alpha \mid \alpha \in \kappa\rangle$ such that

$$\big|[\xi_\alpha, \xi_{\alpha+1})\big| = \big|\bigcup\{B_\delta \mid \xi_\alpha \leq \delta < \xi_{\alpha+1}\}\big|.$$

Then it suffices to define C by $C_\alpha := \bigcup\{B_\delta \mid \xi_\alpha \leq \delta < \xi_{\alpha+1}\}$ for all $\alpha \in \kappa$ and apply Corollary 2.22 (1).

(2). Suppose that $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$, and without loss assume that $|B_\delta|$ is infinite for every $\delta \in \kappa$. Let $\langle\xi_\alpha \mid \alpha \in \kappa\rangle$ be a sequence as defined in (1) above. We define C as follows.

If $\alpha \in \kappa$ is an even ordinal, then we let $C_\alpha := \bigcup\{B_\delta \mid \xi_\alpha \leq \delta < \xi_{\alpha+1}\}$. For each odd ordinal $\gamma \in \kappa$ and δ such that $\xi_\gamma \leq \delta < \xi_{\gamma+1}$, let B'_δ be a partition of B_δ into $|B_\delta|$-many parts of size $|B_\delta|$. We then let $\{C_\alpha \mid \alpha \in \kappa \text{ is odd}\}$ be an enumeration of $\bigcup\{B'_\delta \mid \exists\gamma \in \kappa\,(\gamma \text{ is odd and } \xi_\gamma \leq \delta < \xi_{\gamma+1})\}$. We may again apply Corollary 2.22 (1) to see that $\mathcal{E}_\kappa(B) \nsubseteq \mathcal{E}_\kappa(C) \nsubseteq \mathcal{E}_\kappa(B)$.

(3). The argument is similar to (2) above, but we have to take a little more care in order to deal with intervals. Given $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$, we define $J \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ recursively by iterating through three stages. At each step of the iteration, suppose we have define J_ξ for all $\xi < \alpha$ and that α' is such that $\bigcup_{\xi\in\alpha} J_\xi = \bigcup_{\xi\in\alpha'} I_\xi$. In the first stage, we then define $J_\alpha = \bigcup\{I_\delta \mid \alpha' \leq \delta < \alpha''\}$, where α'' is chosen large enough such that $|J_\alpha| = \big|[\alpha', \alpha'')\big|$. In the second stage, we define $J_{\alpha+1} = [\min(I_{\alpha''}), \min(I_\delta))$, with $\delta > \alpha''$ large enough such that $|I_\xi| > |\alpha|$ for all $\xi \geq \delta$. Finally, in the third stage, we define $J_{\alpha+2+\xi}$ for each $\xi < |I_\delta|$ to be intervals of length α, such that $\bigcup_{\xi<|I_\delta|} J_{\alpha+2+\xi}$ covers the initial segment of I_δ of length $|I_\delta|$. If at this point $\bigcup_{\xi\in|I_\delta|} J_{\alpha+2+\xi} \neq I_\delta$, then we define $J_{\alpha+|I_\delta|}$ as the remaining part of the interval I_δ. This concludes the third stage, and we continue by iteration until J_α has been defined for all $\alpha \in \kappa$.

(4). Let $\kappa = \lambda^+$ and define $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ by $I_\alpha = [\lambda\alpha, \lambda\alpha + \lambda)$ for each $\alpha \in \kappa$. Now, if $J \in \mathrm{IP}^{\mathrm{lim}}_\kappa$, then for almost all $\delta \in \kappa$ we have $|J_\delta| = \lambda$, and thus we see that for almost all $\alpha \in \kappa$ there is a $\delta \in \kappa$ such that $I_\alpha \subseteq J_\delta \cup J_{\delta+1}$. By Lemma 2.21, $\mathcal{E}_\kappa(J) \subseteq \mathcal{E}_\kappa(I)$. □

Corollary 2.25. If κ is successor, then let $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ be such that $\mathcal{E}_\kappa(J) \subseteq \mathcal{E}_\kappa(I)$ for all $J \in \mathrm{IP}^{\mathrm{lim}}_\kappa$, then $\mathcal{E}_\kappa(I) = \mathcal{E}^\exists_\kappa = \mathcal{E}_\kappa$. Particularly, $\mathcal{E}^\exists_\kappa$ is $\leq\kappa$-complete ideal when κ is successor.

Corollary 2.26. Let κ be regular uncountable. Then $\mathcal{E}^\forall_\kappa \subsetneq \mathcal{E}_\kappa(I)$ for all $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$, and $\mathcal{BE}^\forall_\kappa \subsetneq \mathcal{E}_\kappa(B)$ for all $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$.

Proof. Fix some $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$, then Proposition 2.24 (1) shows that $\mathcal{E}_\kappa(I)$ is not least, thus there is $J \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ with $\mathcal{E}_\kappa(I) \nsubseteq \mathcal{E}_\kappa(J)$. It follows from this that $\mathcal{E}^\forall_\kappa \subseteq \mathcal{E}_\kappa(I) \cap \mathcal{E}_\kappa(J) \subsetneq \mathcal{E}_\kappa(I)$. The same argument shows that $\mathcal{BE}^\forall_\kappa \subsetneq \mathcal{E}_\kappa(B)$ for all $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$. □

We will use the characterisation given in Corollary 2.22 to show that $\mathcal{BE}^\exists_\kappa$ is not an ideal.

Lemma 2.27. Suppose that

(1) $B, B' \in \mathrm{BP}^{\lim}_\kappa$ are such that $|B_\alpha \cap B'_\beta| \leq 1$ for almost all $\alpha, \beta \in \kappa$,
(2) $\varphi \in \Sigma_\kappa(B)$ and $\varphi' \in \Sigma_\kappa(B')$ have the property that for any $\alpha, \alpha' \in \kappa$, any $\xi \in B_\alpha \cap B'_{\alpha'}$ and any $i \in \{0,1\}$, there are $s \in \varphi(\alpha)$ and $s' \in \varphi'(\alpha')$ such that $s(\xi) = s'(\xi) = i$, and
(3) $C \in \mathrm{BP}^{\lim}_\kappa$ and $\psi \in \Sigma_\kappa(C)$,

then $[\psi]_* \cap [\varphi]_0 \in \mathcal{M}_\kappa([\varphi]_0)$ or $[\psi]_* \cap [\varphi']_0 \in \mathcal{M}_\kappa([\varphi']_0)$.[1]

Proof. Let $S_\alpha := \{\delta \in \kappa \mid B_\delta \cap C_\alpha \neq \varnothing\}$ and $S'_\alpha := \{\delta \in \kappa \mid B'_\delta \cap C_\alpha \neq \varnothing\}$, then we first show that for almost all $\alpha \in \kappa$ we have $|S_\alpha| = |C_\alpha|$ or $|S'_\alpha| = |C_\alpha|$. Namely, if $\alpha \in \kappa$ and $|S_\alpha| < |C_\alpha|$, then we see that

$$|C_\alpha| = \left|\bigcup_{\delta \in S_\alpha} B_\delta \cap C_\alpha\right| = |S_\alpha| \cdot \sup_{\delta \in S_\alpha} |B_\delta \cap C_\alpha| = \sup_{\delta \in S_\alpha} |B_\delta \cap C_\alpha|.$$

If λ is a cardinal with $\lambda < |C_\alpha|$, then there is $\delta \in S_\alpha$ such that $\lambda \leq |B_\delta \cap C_\alpha|$. Thus, if α is sufficiently large, there are distinct ε_ξ for each $\xi < \lambda$ for which $|B'_{\varepsilon_\xi} \cap B_\delta \cap C_\alpha| = 1$. Hence $\varepsilon_\xi \in S'_\alpha$ for each $\xi < \lambda$, showing that $\lambda \leq |S'_\alpha|$. Since λ was arbitrary, it follows that $|C_\alpha| = |S'_\alpha|$.

Now assume that $\left|\{\alpha \in \kappa \mid |C_\alpha| = |S_\alpha|\}\right| = \kappa$ (the alternative with S'_α in place of S_α is analogous), then we show that $[\psi]_\beta \cap [\varphi]_0$ is nowhere dense in the space $[\varphi]_0$ for each $\beta \in \kappa$. This suffices to prove the lemma, because $[\psi]_* = \bigcup_{\beta \in \kappa} [\psi]_\beta$. Fix some arbitrary $s \in {}^{<\kappa}2$ such that $[s] \cap [\varphi]_0 \neq \varnothing$. Let $\alpha_0 \in \kappa$ be large enough such that

(i) for any $\alpha, \beta \in \kappa$ with $|B_\alpha \cap B'_\beta| > 1$ we have $\sup(B_\alpha \cup B'_\beta) < \alpha_0$,
(ii) $\sup\left(\bigcup_{\xi \in \beta} C_\xi\right) < \alpha_0$, and
(iii) if $\alpha \in \kappa$ is such that $\mathrm{dom}(s) \cap B_\alpha \neq \varnothing$, then $\sup(B_\alpha) < \alpha_0$.

Now we fix $\alpha \geq \alpha_0$ such that $|C_\alpha| = |S_\alpha|$. Since $|\psi(\alpha)| < 2^{|C_\alpha|}$, we may fix for each $\delta \in S_\alpha$ some distinct $\varepsilon_\delta \in B_\delta \cap C_\alpha$, and some $t \colon \{\varepsilon_\delta \mid \delta \in S_\alpha\} \to 2$ such that no extension of t is contained in $\psi(\alpha)$. By (ii), we then see that $[t] \cap [\psi]_\beta = \varnothing$. For each $\delta \in S_\alpha$, fix $t_\delta \in \varphi(\delta)$ such that $t_\delta(\varepsilon_\delta) = t(\varepsilon_\delta)$, which is possible by assumption (2) made in the lemma statement. By (iii), we see that $s' := s \cup \bigcup_{\delta \in S_\alpha} t_\delta$ is a well-defined function and that $[s'] \cap [\varphi]_0 \neq \varnothing$. This proves that $[\psi]_\beta$ is nowhere dense in $[\varphi]_0$. □

Theorem 2.28. Let κ be regular uncountable.

(1) There exist $B, B' \in \mathrm{BP}^{\lim}_\kappa$ such that for all $C \in \mathrm{BP}_\kappa$ we have $\mathcal{E}_\kappa(B) \cup \mathcal{E}_\kappa(B') \not\subseteq \mathcal{E}_\kappa(C)$.
(2) There exist $B, B' \in \mathrm{BP}^{\lim}_\kappa$, $\varphi \in \Sigma_\kappa(B)$ and $\varphi' \in \Sigma_\kappa(B')$ such that $[\varphi]_* \cup [\varphi']_* \notin \mathcal{BE}^{\exists}_\kappa$.

Proof. (1). Let $B, B' \in \mathrm{BP}^{\lim}_\kappa$ be as in Lemma 2.27, and define $S_\alpha = \{\delta \in \kappa \mid B_\delta \cap C_\alpha \neq \varnothing\}$ and $S'_\alpha = \{\delta \in \kappa \mid B'_\delta \cap C_\alpha \neq \varnothing\}$. Then, as in the proof of Lemma 2.27, we see that for any $C \in \mathrm{BP}^{\lim}_\kappa$, for almost every $\alpha \in \kappa$ one of $|S_\alpha| = |C_\alpha|$ or $|S'_\alpha| = |C_\alpha|$ holds. Therefore, by Corollary 2.22, we have $\mathcal{E}_\kappa(B) \not\subseteq \mathcal{E}_\kappa(C)$ or $\mathcal{E}_\kappa(B') \not\subseteq \mathcal{E}_\kappa(C)$.

(2). Let $B, B' \in \mathrm{BP}^{\lim}_\kappa$, $\varphi \in \Sigma_\kappa(B)$ and $\varphi' \in \Sigma_\kappa(B')$ be as in Lemma 2.27. For any $C \in \mathrm{BP}^{\lim}_\kappa$ and $\psi \in \Sigma_\kappa(C)$, Lemma 2.27 tells us that $[\psi]_* \cap [\varphi]_0 \in \mathcal{M}_\kappa([\varphi]_0)$ or $[\psi]_* \cap [\varphi']_0 \in \mathcal{M}_\kappa([\varphi']_0)$. This obviously implies that $[\varphi]_* \cup [\varphi']_* \not\subseteq [\psi]_*$. It follows that $[\varphi]_* \cup [\varphi']_* \notin \mathcal{BE}^{\exists}_\kappa$. □

With a slightly different argument we can show that $\mathcal{E}^{\exists}_\kappa$ is not an ideal either under certain assumptions, including when κ is strongly inaccessible, but excluding the successor case given by Corollary 2.25. The following lemma is the interval version of Lemma 2.27.

Lemma 2.29. Suppose that

[1] This lemma originally only showed the existence of such φ, φ' for which $[\varphi]_* \cup [\varphi']_* \notin \mathcal{BE}^{\exists}_\kappa$, and was sharpened to its current form by ChatGPT 5.6. The proof presented here, however, was written by the authors.

(1) $\langle L_\alpha \mid \alpha \in \kappa\rangle \in \mathrm{IP}_\kappa^{\lim}$ is such that $2^{|L_\alpha|} < 2^{|L_\beta|}$ for any $\alpha < \beta$,
(2) $x, x' \in {}^\kappa\kappa$ are strictly increasing such that $|\mathrm{ran}(x) \cap \mathrm{ran}(x')| < \kappa$,
(3) $I, I' \in \mathrm{IP}_\kappa^{\lim}$ have $\sup(I_\alpha) = \sup(L_{x(\alpha)+1})$ and $\sup(I'_\alpha) = \sup(L_{x'(\alpha)+1})$ for each $\alpha \in \kappa$,
(4) $\varphi \in \Sigma_\kappa(I)$ and $\varphi' \in \Sigma_\kappa(I')$ are such that
  (a) each $s\colon I_\alpha \setminus L_{x(\alpha)+1} \to 2$ has some $t \in \varphi(\alpha)$ such that $s \subseteq t$, and
  (b) each $s'\colon I'_\alpha \setminus L_{x'(\alpha)+1} \to 2$ has some $t' \in \varphi'(\alpha)$ such that $s' \subseteq t'$,
  (c) $2 \leq \min\{|\varphi(\alpha)|, |\varphi'(\alpha)|\}$ for all $\alpha \in \kappa$,
(5) $B \in \mathrm{BP}_\kappa$ and $\psi \in \Sigma_\kappa(B)$,

then $[\psi]_* \cap [\varphi]_0 \in \mathcal{M}_\kappa([\varphi]_0)$ or $[\psi]_* \cap [\varphi']_0 \in \mathcal{M}_\kappa([\varphi']_0)$.

Before proving Lemma 2.29, let us make some remarks on the consistency of its assumptions. Clearly, a sequence as in (1) exists for strongly inaccessible κ, but it also may exist for weakly inaccessible κ. The situation in which such a sequence exist is useful enough to give a name.

Definition 2.30. Define the *continuum function* on cardinal numbers as the function $\lambda \mapsto 2^\lambda$, and say it is *eventually constant* below κ if for some $\lambda < \kappa$ we have $2^\lambda = 2^{\lambda'}$ for all cardinals $\lambda' \in [\lambda, \kappa)$.

Observe that the continuum function is "eventually" constant below any successor cardinal, and is not eventually constant below any strongly inaccessible cardinal, although it is independent for weakly inaccessible cardinals.

We also note that (4) can be satisfied if a sequence as in (1) exists: if $\langle L_\alpha \mid \alpha \in \kappa\rangle$ is as in (1), then $2^{|\bigcup_{\xi\leq\alpha} L_\alpha|} < 2^{|L_{\alpha+1}|}$; it follows that the cardinality of the set of all $s\colon I_\alpha \setminus L_{x(\alpha)+1} \to 2$ equals $2^{|L_{x(\alpha)}|}$, which is strictly smaller than $2^{|I_\alpha|} = 2^{|L_{x(\alpha)+1}|}$, and thus φ as in (4) can be found.

Proof of Lemma 2.29. Let $L := \bigcup_{\xi\in\kappa}(I_\xi \setminus L_{x(\xi)+1})$ and $L' := \bigcup_{\xi\in\kappa}(I'_\xi \setminus L_{x'(\xi)+1})$, then (2) implies $|\kappa \setminus (L \cup L')| < \kappa$. Therefore, there are either cofinally many $\alpha \in \kappa$ for which $|B_\alpha \cap L| = |B_\alpha|$ holds, or for which $|B_\alpha \cap L'| = |B_\alpha|$ holds. Let us assume the former, as the two options are symmetric. We then show that $[\psi]_\beta \cap [\varphi]_0$ is nowhere dense in $[\varphi]_0$.

Fix $s \in {}^{<\kappa}2$ with $[s] \cap [\varphi]_0 \neq \varnothing$ and choose $\alpha \in \kappa$ such that

(i) $|B_\alpha \cap L| = |B_\alpha|$,
(ii) If $\xi \in \kappa$ is such that $I_\xi \cap B_\alpha \neq \varnothing$, then $\mathrm{dom}(s) < \min(I_\xi)$.

Since $|\psi(\alpha)| < 2^{|B_\alpha|}$, we may choose $t\colon B_\alpha \cap L \to 2$ such that no extension of t is in $\psi(\alpha)$. For each $\xi \in \kappa$ such that $I_\xi \cap B_\alpha \neq \varnothing$, let $t_\xi = t \restriction I_\xi$, then assumption (4) implies that there exists $t'_\xi \in \varphi(\xi)$ such that $t_\xi \subseteq t'_\xi$. Moreover, (ii) implies that $s' := s \cup \bigcup\{t'_\xi \mid \xi \in \kappa \text{ and } I_\xi \cap B_\alpha \neq \varnothing\}$ is a well-defined function, and we see that $[s'] \cap [\varphi]_0 \neq \varnothing$. Since $t \subseteq s'$ we have $[s'] \cap [\psi]_\beta = \varnothing$. Finally, s and β were arbitrary, so $[\psi]_* \cap [\varphi]_0 \in \mathcal{M}_\kappa([\varphi]_0)$. □

Theorem 2.31. The following are equivalent for regular uncountable κ.

(1) The continuum function is eventually constant below κ.
(2) The family $\mathcal{E}_\kappa^\exists$ is closed under unions.
(3) For every $B \in \mathrm{BP}_\kappa^{\lim}$ there exists $I \in \mathrm{IP}_\kappa^{\lim}$ with $\mathcal{E}_\kappa(I) \subseteq \mathcal{E}_\kappa(B)$.

Proof. (1) ⇒ (2). If κ is successor, then this follows from Corollary 2.25, so let us assume that κ is weakly inaccessible, and that $\lambda < \kappa$ is such that $2^{\lambda'} = 2^\lambda$ for all cardinals $\lambda' \in [\lambda, \kappa)$. Suppose that $I, I' \in \mathrm{IP}_\kappa^{\lim}$ and $\varphi \in \Sigma_\kappa(I)$ and $\varphi' \in \Sigma_\kappa(I')$. Let $J = \langle J_\alpha \mid \alpha \in \kappa\rangle$ be the enumeration the set $\{I_\alpha \cap I'_{\alpha'} \mid \alpha, \alpha' \in \kappa\} \setminus \{\varnothing\}$ such that $J \in \mathrm{IP}_\kappa$. We don't necessarily have $J \in \mathrm{IP}_\kappa^{\lim}$; however, it is easy

to see that $\lim_{\alpha\to\kappa} |J_\alpha \cup J_{\alpha+1}| = \kappa$, and thus we may find $J' \in \mathrm{IP}^{\lim}_\kappa$ such that for each $\alpha' \in \kappa$, there is $\alpha \in \kappa$ such that $J'_{\alpha'} = J_\alpha$ or $J'_{\alpha'} = J_\alpha \cup J_{\alpha+1}$.

Now define $\psi \in \Sigma_\kappa(J')$ as follows. Given J'_α, either there is one or there are two $\xi \in \kappa$ such that $J'_\alpha \cap I_\xi \neq \varnothing$. For any such ξ, let $\{s \restriction J'_\alpha \mid s \in \varphi(\xi)\} \subseteq \psi(\alpha)$. We similarly incorporate $\varphi'(\xi')$ into $\psi(\alpha)$ for the relevant one or two ξ'. If α is large enough, then $2^{|J'_\alpha|} = 2^{|I_\xi|}$, and hence $|\varphi(\xi)| < 2^{|J'_\alpha|}$, hence we may find such ψ with $\psi \in \Sigma^*_\kappa(J')$. It follows that $[\varphi]_* \cup [\varphi']_* \subseteq [\psi]_*$.

(2) ⇒ (1). We prove the contrapositive. If the continuum function is not eventually constant below κ, then the assumptions of Lemma 2.29 can be satisfied. The rest follows as in Theorem 2.28 (2), using Lemma 2.29 instead of Lemma 2.27.

(1) ⇒ (3). Assume that the continuum function is eventually constant below κ, and let $\lambda < \kappa$ be such that $2^\lambda = 2^{\lambda'}$ for all $\lambda' \in [\lambda, \kappa)$. Given $B \in \mathrm{BP}^{\lim}_\kappa$, we see that $|B_\alpha| \geq \lambda$ for almost all $\alpha \in \kappa$. Let $I \in \mathrm{IP}_\kappa$ be defined as in Lemma 2.13 such that every I_δ has a set $A_\delta \subseteq \kappa$ with $I_\delta = \bigcup_{\alpha\in A_\delta} B_\alpha$, and let δ_α be such that $B_\alpha \subseteq I_{\delta_\alpha}$. Hence, if $S_\alpha := \{\delta \in \kappa \mid I_\delta \cap B_\alpha \neq \varnothing\}$, then $S_\alpha = \{\delta_\alpha\}$. Note that $I \in \mathrm{IP}^{\lim}_\kappa$, and thus for large enough α we see that $|I_{\delta_\alpha}| \geq \lambda$, which implies that $2^{|I_{\delta_\alpha}|} = 2^{|B_\alpha|}$ for almost all $\alpha \in \kappa$. Therefore, for any $\langle \kappa_{\alpha,\delta} \mid \alpha \in \kappa, \delta \in S_\alpha\rangle$ such that $\kappa_{\alpha,\delta} < 2^{|I_\delta|}$, we see that, for almost all $\alpha \in \kappa$,

$$\prod_{\delta\in S_\alpha} \kappa_{\alpha,\delta} = \kappa_{\alpha,\delta_\alpha} < 2^{|I_{\delta_\alpha}|} = 2^{|B_\alpha|}.$$

Thus, by Lemma 2.21 it follows that $\mathcal{E}_\kappa(I) \subseteq \mathcal{E}_\kappa(B)$.

(3) ⇒ (1). Suppose $\langle \lambda_\alpha \mid \alpha \in \kappa\rangle$ is a sequence of infinite cardinals below κ such that $2^{\lambda_\alpha} < 2^{\lambda_{\alpha'}}$ for each $\alpha < \alpha'$. We define disjoint sets $B_\alpha \subseteq \kappa$ such that for each $\alpha \in \kappa$ we have $|B_\alpha| = \lambda_\alpha$, but such that for any $\zeta \in B_\alpha$ we have $[\zeta, \zeta + \lambda_{\alpha+1}) \cap B_\alpha = \{\zeta\}$. It is not hard to see that we can additionally require $\bigcup B_\alpha = \kappa$, meaning that $B := \langle B_\alpha \mid \alpha \in \kappa\rangle \in \mathrm{BP}^{\lim}_\kappa$. Let $I \in \mathrm{IP}^{\lim}_\kappa$ and $\delta \in \kappa$. If $|I_\delta| < \lambda_{\alpha+1}$, then $2^{|I_\delta \cap B_\alpha|} \leq 2^1 < 2^{|I_\delta|}$, and otherwise we have $2^{|I_\delta\cap B_\alpha|} \leq 2^{|B_\alpha|} = 2^{\lambda_\alpha} < 2^{|I_\delta|}$. By Corollary 2.22 (2), we see that $\mathcal{E}_\kappa(I) \not\subseteq \mathcal{E}_\kappa(B)$. □

Corollary 2.32. If κ is regular uncountable and the continuum function is not eventually constant below κ, then $\mathcal{E}_\kappa \not\subseteq \mathcal{BE}^\exists_\kappa$.

Proof. This holds, because the set $[\varphi]_* \cup [\varphi']_*$ constructed in Theorem 2.31 (2) ⇒ (1) is contained in $\mathcal{E}_\kappa \setminus \mathcal{BE}^\exists_\kappa$. □

Finally, Theorem 2.31 allows us to conclude something about the inclusion $\mathcal{BE}^\forall_\kappa \subseteq \mathcal{E}^\forall_\kappa$.

Theorem 2.33. If the continuum function is eventually constant below κ, then $\mathcal{E}^\forall_\kappa = \mathcal{BE}^\forall_\kappa$.

Proof. This follows Theorem 2.31 (1) ⇒ (3) and the definitions of $\mathcal{E}^\forall_\kappa$ and $\mathcal{BE}^\forall_\kappa$. □

3. Additive ideals

In this section we will study additive closure of the ideals from Definition 2.7.

3.1. **Definitions of additive ideals and related ideals.** For $x, y \in {}^\kappa 2$, let us define addition on ${}^\kappa 2$ by the sum $x + y$ being the function that maps $\alpha \mapsto 0$ if $x(\alpha) = y(\alpha)$ and $\alpha \mapsto 1$ otherwise. We expand this to subsets of $X, Y \subseteq {}^\kappa 2$ by defining

$$x + Y = \{x + y \mid y \in Y\}, \qquad X + y = \{x + y \mid x \in X\},$$
$$X + Y = \{x + y \mid x \in X \text{ and } y \in Y\}.$$

For an ideal $\mathcal{I}_\kappa$ on ${}^\kappa 2$, we call a set $X \subseteq {}^\kappa 2$ an *$\mathcal{I}_\kappa$-additive set* if the sum $X + Y$ is in $\mathcal{I}_\kappa$ for all $Y \in \mathcal{I}_\kappa$. The family of $\mathcal{I}_\kappa$-additive sets is an ideal, which we will denote by $\mathcal{IA}_\kappa$. We call an ideal that is closed under addition an *additive ideal*. We make some easy observations.

Proposition 3.1.

(1) $\mathcal{IA}_\kappa \subseteq \mathcal{I}_\kappa$,
(2) $\mathcal{IA}\mathcal{A}_\kappa = \mathcal{IA}_\kappa$, that is, the familly of $\mathcal{IA}_\kappa$-additive sets is simply $\mathcal{IA}_\kappa$,
(3) If $\mathcal{I}_\kappa$ is $<\lambda$-complete, then so is $\mathcal{IA}_\kappa$,
(4) If $X + Y \in \mathcal{I}_\kappa$ for all $X, Y \in \mathcal{I}_\kappa$, then $\mathcal{I}_\kappa = \mathcal{IA}_\kappa$.

In this section, we will study the additive counterparts to the ideals defined in Definition 2.7. Before we do so, we will need two additional ideals.

Definition 3.2. A set $X \subseteq {}^\kappa 2$ is *strongly κ-null* if for every $A \in [\kappa]^\kappa$ there exists $s_\alpha \in {}^\alpha 2$ for every $\alpha \in A$ such that $X \subseteq \bigcup_{\alpha\in A}[s_\alpha]$. We write $\mathcal{SN}_\kappa$ for the ideal of strongly κ-null subsets of ${}^\kappa 2$.

The above definition is a straightforward generalisation of the strongly null ideal on ${}^\omega 2$, or of the ideal of strong measure zero sets on the reals. The κ-version of this ideal has been previously considered in [18, 19, 11].

Definition 3.3. A set $X \subseteq {}^\kappa 2$ is *κ-null-additive* if for every $I \in \mathrm{IP}_\kappa$ there exists $\varphi \in \prod_{\alpha\in\kappa}\left[{}^{I_\alpha}2\right]^{|\alpha|}$ such that $X \subseteq [\varphi]_*$. We write $\mathcal{NA}_\kappa$ for the ideal of κ-null-additive subsets of ${}^\kappa 2$.

As the name suggests, classically the property described above is equivalent to Lebesgue measure null-additive subsets of ${}^\omega 2$, as was shown by Shelah [28]. The above characterisation was used to show that Lebesgue null-additive sets are meagre-additive, and the proof generalises without any hardship.

Theorem 3.4 (cf. [28])**.** $\mathcal{NA}_\kappa \subseteq \mathcal{MA}_\kappa$. □

The Galvin–Mycielski–Solovay theorem [15, 16] connects meagre-additive sets to the strongly null ideal. It generalises to the higher context for strongly inaccessible κ:

Theorem 3.5 (Wohofsky [32], Schürz [27])**.** If a set $X \subseteq {}^\kappa 2$ has the property that for every $M \in \mathcal{M}_\kappa$ we have $X + M \neq {}^\kappa 2$, then $X \in \mathcal{SN}_\kappa$. If κ is strongly inaccessible, then the reverse holds as well. □

Corollary 3.6. $\mathcal{MA}_\kappa \subseteq \mathcal{SN}_\kappa$.

3.2. **Higher closed null-additive.** Note that the definition of $\mathcal{NA}_\kappa$ essentially involves an intersection of a family of ideals based on slaloms on interval partitions, so it should come as no surprise that $\mathcal{NA}_\kappa$ is related to the ideals $\mathcal{E}^\forall_\kappa$ and $\mathcal{BE}^\forall_\kappa$. Specifically, we will show that $\mathcal{NA}_\kappa \subseteq \mathcal{BE}^\forall_\kappa$. Before we do so, we will prove the weaker claim that $\mathcal{NA}_\kappa \subseteq \mathcal{E}^\forall_\kappa$ and use that, for successor κ, $\mathcal{E}^\forall_\kappa = \mathcal{BE}^\forall_\kappa$ by Theorem 2.33.

Lemma 3.7. Let κ be regular uncountable, then $\mathcal{NA}_\kappa \subseteq \mathcal{E}^\forall_\kappa$.

Proof. Let $A \in \mathcal{NA}_\kappa$ and fix some arbitrary $I \in \mathrm{IP}^{\lim}_\kappa$. We will find $\varphi \in \Sigma_\kappa(I)$ such that $A \subseteq [\varphi]_*$.

We inductively define a continuous increasing sequence $\langle \alpha_\xi \mid \xi \in \kappa\rangle$ of ordinals in κ such that for all $\xi \in \kappa$ and $\alpha \geq \alpha_\xi$ we have $|\xi| \leq |I_\alpha|$. Define $J \in \mathrm{IP}^{\lim}_\kappa$ by $J_\xi = \bigcup\{I_\alpha \mid \alpha_\xi \leq \alpha < \alpha_{\xi+1}\}$ for each $\xi \in \kappa$. Since $A \in \mathcal{NA}_\kappa$, there is $\psi \in \prod_{\xi\in\kappa}\left[{}^{J_\xi}2\right]^{|\xi|}$ such that $A \subseteq [\psi]_*$. Let φ be defined such that for any $\xi, \alpha \in \kappa$ with $\alpha \in [\alpha_\xi, \alpha_{\xi+1})$ we have $\varphi : \alpha \mapsto \{s \restriction I_\alpha \mid s \in \psi(\xi)\}$. Note that

$$|\varphi(\alpha)| \leq |\psi(\xi)| \leq |\xi| \leq |I_\alpha| < 2^{|I_\alpha|}.$$

Therefore $\varphi \in \Sigma_\kappa(I)$, and since $[\psi]_* \subseteq [\varphi]_*$ it follows that $A \in \mathcal{E}_\kappa(I)$. Finally, as I was arbitrary, we see that $\mathcal{NA}_\kappa \subseteq \mathcal{E}^\forall_\kappa$. □

Theorem 3.8. Let κ be regular uncountable, then $\mathcal{NA}_\kappa \subseteq \mathcal{BE}^\forall_\kappa$.

Proof. If κ is successor, then by Lemma 3.7 and Theorem 2.33 it follows that $\mathcal{NA}_\kappa \subseteq \mathcal{BE}^\forall_\kappa$. Let us therefore assume that κ is regular limit. Let $A \in \mathcal{NA}_\kappa$ and fix some arbitrary $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$, then we will find $\varphi \in \Sigma_\kappa(B)$ such that $A \subseteq [\varphi]_*$.

Since $\lim_{\beta\in\kappa}|B_\beta| = \kappa$ is limit, we may define $\beta_\xi = \sup\big(\bigcup\{B_\beta \mid \beta \in \kappa \text{ and } |B_\beta| \leq |\xi|\}\big)$.[2] Let $f \in {}^\kappa\kappa$ be defined by

$$f : \alpha \mapsto \min\big\{\gamma \geq \alpha \;\big|\; \forall\beta \in \kappa\,(B_\beta \cap \gamma \neq \varnothing \to B_\beta \subseteq \gamma)\big\}.$$

By recursion on $\alpha \in \kappa$, we define a continuous strictly increasing sequence of ordinals $\langle\gamma_\alpha \mid \alpha \in \kappa\rangle$ such that $\gamma_0 = 0$ and $\gamma_{\alpha+1} = f(\max\{\gamma_\alpha, \beta_\alpha\} + 1)$. Let $J \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ be defined by $J_\alpha = [\gamma_\alpha, \gamma_{\alpha+1})$.

Since $A \in \mathcal{NA}_\kappa$, there exists $\psi \in \prod_{\alpha\in\kappa}\big[{}^{J_\alpha}2\big]^{|\alpha|}$ such that $A \subseteq [\psi]_*$. Now define φ such that for any $\alpha, \beta \in \kappa$ with $B_\beta \subseteq J_\alpha$ we have $\varphi\colon \alpha \mapsto \{s \restriction B_\beta \mid s \in \psi(\alpha)\}$. Since $B_\beta \subseteq J_\alpha$ implies that for each $\xi \in \alpha$ we have $B_\beta \cap \beta_\xi = \varnothing$, it follows from the construction of β_ξ that $|\alpha| \leq |B_\beta|$. In particular, if $B_\beta \subseteq J_\alpha$, then

$$|\varphi(\beta)| \leq |\psi(\alpha)| \leq |\alpha| \leq |B_\beta| < 2^{|B_\beta|}.$$

It follows that $\varphi \in \Sigma_\kappa(B)$ and $[\psi]_* \subseteq [\varphi]_*$, so $A \in \mathcal{E}_\kappa(B)$ and since B was arbitrary $\mathcal{NA}_\kappa \subseteq \mathcal{BE}^\forall_\kappa$. □

Theorem 3.9. Let κ be regular uncountable, then $\mathcal{E}^\forall_\kappa \subseteq \mathcal{EA}_\kappa \subseteq \mathcal{E}_\kappa$.

Proof. That $\mathcal{EA}_\kappa \subseteq \mathcal{E}_\kappa$ is clear.

Assume $X \in \mathcal{E}^\forall_\kappa$ and for each $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ pick some $\psi^I \in \Sigma_\kappa(I)$ such that $X \subseteq \bigcap_{I\in\mathrm{IP}^{\mathrm{lim}}_\kappa}\big[\psi^I\big]_*$. Let $E \in \mathcal{E}_\kappa$ be arbitrary, then there is a sequence $\langle I_\beta \mid \beta \in \kappa\rangle$ in $\mathrm{IP}^{\mathrm{lim}}_\kappa$ and $\varphi^\beta \in \Sigma_\kappa(I_\beta)$ for each $\beta \in \kappa$ such that $E \subseteq \bigcup_{\beta\in\kappa}\big[\varphi^\beta\big]_*$. Given $\beta \in \kappa$, let us define $\chi^\beta : \alpha \mapsto \varphi^\beta(\alpha) + \psi^{I_\beta}(\alpha)$, and note that $\chi^\beta \in \Sigma_\kappa(I_\beta)$. Finally note that

$$E + X \subseteq \bigcup_{\beta\in\kappa}\big[\varphi^\beta\big]_* + \bigcap_{I\in\mathrm{IP}^{\mathrm{lim}}_\kappa}\big[\psi^I\big]_* \subseteq \bigcup_{\beta\in\kappa}\big(\big[\varphi^\beta\big]_* + \big[\psi^{I_\beta}\big]\big) = \bigcup_{\beta\in\kappa}\big[\chi^\beta\big]_* \in \mathcal{E}_\kappa.$$ □

Both of the inclusions given in Theorem 3.9 are strict for certain choices of κ, albeit for different choices. If κ is strongly inaccessible, then we will see that $\mathcal{EA}_\kappa \subsetneq \mathcal{E}_\kappa$ as a consequence of Theorem 3.11. The case for successor κ is different. Note that ideals of the form $\mathcal{E}_\kappa(B)$ for $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$ are additively closed (which also implies that $\mathcal{E}^\forall_\kappa$ and $\mathcal{BE}^\forall_\kappa$ are additive ideals).

Lemma 3.10. If κ is successor, then $\mathcal{E}^\forall_\kappa \subsetneq \mathcal{EA}_\kappa = \mathcal{E}_\kappa$.

Proof. By Corollary 2.25 we may choose $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ with $\mathcal{E}_\kappa = \mathcal{E}_\kappa(I)$, and since $\mathcal{E}_\kappa(I)$ is additive, we see that $\mathcal{E}_\kappa = \mathcal{EA}_\kappa$. It then follows from Corollary 2.26 that $\mathcal{E}^\forall_\kappa \subsetneq \mathcal{E}_\kappa(I) = \mathcal{E}_\kappa$. □

3.3. **Decomposing the higher Cantor space.** In this subsection we study some cases where additivity fails, and where large sets can be decomposed into the sum of small sets. As mentioned above, $\mathcal{E}_\kappa(B)$ is an additive ideal for any $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$. We will show that $\mathcal{E}^\exists_\kappa$ does not have to be additive (and thus neither does $\mathcal{E}_\kappa$), and $\mathcal{BE}^\exists_\kappa$ is not additive for any regular uncountable κ.

Theorem 3.11. Let κ be regular limit, then the following are equivalent:

(1) there exist $X, Y \in \mathcal{E}^\exists_\kappa$ for which $X + Y = {}^\kappa 2$.
(2) the continuum function is not eventually constant below κ,

[2] We assume that $\sup(\varnothing) = 0$.

Proof. We first assume that the continuum function is not eventually constant. Let $\langle \kappa_\alpha \mid \alpha \in \kappa \rangle$ be a sequence of cardinals such that $2^{\kappa_\alpha} < 2^{\kappa_\beta}$ for all $\alpha < \beta$. We define $I, J \in \mathrm{IP}_\kappa^{\lim}$, where for each $\alpha \in \kappa$ we have

$$J_\alpha = [\kappa_{2\alpha}, \kappa_{2\alpha+2}), \qquad I_\alpha = \begin{cases} [\kappa_{2\alpha}, \kappa_{2\alpha+1}) & \text{if } \alpha \text{ is limit,} \\ [\kappa_{2\alpha+1}, \kappa_{2\alpha+3}) & \text{otherwise.} \end{cases}$$

Now we define the sets

$$X := \big\{ x \in {}^\kappa 2 \mid \forall \alpha \in \kappa \, \forall \xi \in [\kappa_{2\alpha}, \kappa_{2\alpha+1}) \, (x(\xi) = 0) \big\},$$
$$Y := \big\{ y \in {}^\kappa 2 \mid \forall \alpha \in \kappa \, \forall \xi \in [\kappa_{2\alpha+1}, \kappa_{2\alpha+2}) \, (y(\xi) = 0) \big\}.$$

Then $X \in \mathcal{E}_\kappa(I)$ and $Y \in \mathcal{E}_\kappa(J)$, and $X + Y = {}^\kappa 2$.

Now let us assume that the continuum function is eventually constant, and fix $\lambda < \kappa$ such that $2^{\lambda'} \leq 2^\lambda$ for all $\lambda' < \kappa$. Let $X \in \mathcal{E}_\kappa(I)$ and $Y \in \mathcal{E}_\kappa(J)$ for some $I, J \in \mathrm{IP}_\kappa^{\lim}$, then there exist two strictly increasing sequences $\langle \alpha_\xi \mid \xi \in \kappa \rangle$ and $\langle \beta_\xi \mid \xi \in \kappa \rangle$ such that, for all $\xi \in \kappa$,

$$2^{\left| I_{\alpha_\xi} \right|} = 2^{\left| J_{\beta_\xi} \right|} = 2^{\left| I_{\alpha_\xi} \cap J_{\beta_\xi} \right|} = 2^\lambda.$$

But, since $\left| X \restriction I_{\alpha_\xi} \right| < 2^\lambda$ and $\left| Y \restriction J_{\beta_\xi} \right| < 2^\lambda$, we also see that $(X + Y) \restriction (I_{\alpha_\xi} \cap J_{\beta_\xi}) < 2^\lambda$, and thus $X + Y \neq {}^\kappa 2$. □

Theorem 3.12. Let κ be regular uncountable, then there exist $X, Y \in \mathcal{BE}_\kappa^\exists$ such that $X + Y = {}^\kappa 2$.

Proof. We define $B, C \in \mathrm{BP}_\kappa^{\lim}$ and $\varphi \in \Sigma_\kappa(B)$ and $\psi \in \Sigma_\kappa(C)$ in such a way that for every $\xi, \xi' \in \kappa$ we have $|B_\xi \cap C_{\xi'}| \leq 1$, and there exist unique $b_\xi \in B_\xi$ and $c_{\xi'} \in C_{\xi'}$ such that

(1) $s \in \varphi(\xi)$ if and only if $s \colon B_\xi \to 2$ and $s(\eta) = 0$ for all $\eta \neq b_\xi$,
(2) $t \in \psi(\xi')$ if and only if $t \colon C_{\xi'} \to 2$ and $t(\eta) = 0$ for all $\eta \neq c_{\xi'}$, and
(3) $B_\xi \cap C_{\xi'} \subseteq \{b_\xi, c_{\xi'}\}$.

It is clear that such B, C, φ, ψ can be constructed recursively. Note that $|\varphi(\xi)| = |\psi(\xi')| = 2$ for all $\xi, \xi' \in \kappa$, hence $\varphi \in \Sigma_\kappa(B)$ and $\psi \in \Sigma_\kappa(C)$. We also note that $\kappa = \{b_\xi \mid \xi \in \kappa\} \cup \{c_{\xi'} \mid \xi' \in \kappa\}$, thus if $z \in {}^\kappa 2$ and if we define $z_B, z_C \in {}^\kappa 2$ by

$$z_B \colon \alpha \mapsto \begin{cases} z(\alpha) & \text{if } \alpha \in \{b_\xi \mid \xi \in \kappa\}, \\ 0 & \text{otherwise,} \end{cases} \qquad z_C \colon \alpha \mapsto \begin{cases} z(\alpha) & \text{if } \alpha \notin \{b_\xi \mid \xi \in \kappa\}, \\ 0 & \text{otherwise,} \end{cases}$$

then $z = z_B + z_C$ and $z_B \in [\varphi]_*$ and $z_C \in [\psi]_*$. Thus $[\varphi]_* + [\psi]_* = {}^\kappa 2$. □

Next we will show that $\mathcal{E}_\kappa(B)$ is not κ-meagre-additive.

Theorem 3.13. For every $B \in \mathrm{BP}_\kappa^{\lim}$ there are $X \in \mathcal{E}_\kappa(B)$ and $N \in \mathcal{NWD}$ with $X + N = {}^\kappa 2$.

Proof. Let $\varphi \in \Sigma_\kappa(B)$ be defined by $s \in \varphi(\alpha)$ if and only if $s \colon B_\alpha \to 2$ and $\forall \xi, \xi' \in B_\alpha \, (s(\xi) = s(\xi'))$. Define $N \subseteq {}^\kappa 2$ by $x \in N$ if and only if for all $\alpha \in \kappa$ there are $\xi, \xi' \in B_\alpha$ with $x(\xi) \neq x(\xi')$. It is easy to see that $N \in \mathcal{NWD}$ and $[\varphi]_* + N = {}^\kappa 2$. □

Corollary 3.14. Let κ be regular uncountable and $B \in \mathrm{BP}_\kappa$, then $\mathcal{E}_\kappa(B) \not\subseteq \mathcal{MA}_\kappa$. In particular, $\mathcal{E}_\kappa^\exists \not\subseteq \mathcal{MA}_\kappa$.

Using the Galvin–Mycielski–Solovay characterisation of $\mathcal{SN}_\kappa$ (Theorem 3.5), we may strengthen the above corollary and conclude that $\mathcal{E}_\kappa(B) \not\subseteq \mathcal{SN}_\kappa$ for any $I \in \mathrm{BP}_\kappa^{\lim}$ if κ is strongly inaccessible. On the other hand, it is consistent with κ strongly inaccessible that $\mathcal{E}_\kappa^\forall \subseteq \mathcal{SN}_\kappa$:

Theorem 3.15. Assume GCH. If κ is strongly inaccessible, then $\mathcal{E}_\kappa^\forall \subseteq \mathcal{SN}_\kappa$.

Proof. Fix some $X \in \mathcal{E}_\kappa^\forall$ and let $A \in [\kappa]^\kappa$, then we will first find a suitable $I \in \mathrm{IP}_\kappa^{\lim}$ by inductively defining a continuous strictly increasing sequence of cardinals $\langle \lambda_\alpha \mid \alpha \in \kappa \rangle$ with $\lambda_0 = 0$ and letting $I_\alpha = [\lambda_\alpha, \lambda_{\alpha+1})$. We only need to consider the successor case, so assume we have defined λ_α, then we choose $\lambda_{\alpha+1}$ such that $\lambda_{\alpha+1} = \big|A \cap I_\alpha\big|$; for instance, let $j_0 > \lambda_\alpha$ be large enough such that $\lambda_\alpha^+ \leq \big|A \cap [\lambda_\alpha, j_0)\big|$, for each $n \in \omega$ pick j_{n+1} large enough such that $|j_n| < \big|A \cap [\lambda_\alpha, j_{n+1})\big|$, and let $\lambda_{\alpha+1} = \sup_{n\in\omega} j_n$.

Since $X \in \mathcal{E}_\kappa^\forall$, there exists $\varphi \in \Sigma_\kappa(I)$ such that $X \subseteq [\varphi]_*$. For a given $\alpha \in \kappa$, let us define $X_\alpha := \{x \in X \mid x \restriction I_\alpha \in \varphi(\alpha)\}$. Since $|\varphi(\alpha)| < 2^{|I_\alpha|} = \lambda_{\alpha+1}^+$ and $\big|{}^{\lambda_\alpha}2\big| = \lambda_\alpha^+ \leq \lambda_{\alpha+1}$, we can enumerate $X_\alpha \restriction \lambda_{\alpha+1}$ as $\langle x_\xi \mid \xi \in \lambda_{\alpha+1}\rangle$ and $A \cap I_\alpha$ as $\langle a_\xi^\alpha \mid \xi \in \lambda_{\alpha+1}\rangle$. Then $X_\alpha \subseteq \bigcup_{\xi\in\lambda_{\alpha+1}} [x_\xi \restriction a_\xi^\alpha]$. Finally note that $X = \bigcup_{\alpha\in\kappa} X_\alpha$. Since A was arbitrary, it follows that $X \in \mathcal{SN}_\kappa$. □

We remark that in the classical case $\mathcal{MA} = \mathcal{EA}$, as was shown by Zindulka [35]. For successor κ this cannot be generalised, since successor κ satisfy $\mathcal{EA}_\kappa = \mathcal{E}_\kappa \not\subseteq \mathcal{MA}_\kappa$ by Corollary 3.14.

4. Cardinal characteristics

In this section we will study the cardinal characteristics of the families from Definition 2.7 and compare them to several well-known cardinal characteristics of the higher Baire space. For the ZFC-provable results, it is often possible to provide a Galois–Tukey connection between two relational systems in order to compare cardinal characteristics (cf. [6]). Although the use of relational systems provides more general theorems, we will, for the sake of clarity, give our proofs in a direct fashion, and leave the translation to relational systems to the reader.

Definition 4.1. Let $\mathcal{I}_\kappa$ be a $\leq\kappa$-complete ideal on ${}^\kappa 2$, then we define the following cardinal characteristic parametrised by $\mathcal{I}_\kappa$:

(1) The *additivity number* $\mathrm{add}(\mathcal{I}_\kappa) = \min\big\{|\mathcal{A}| \;\big|\; \mathcal{A} \subseteq \mathcal{I}_\kappa \text{ and } \bigcup \mathcal{A} \notin \mathcal{I}_\kappa\big\}$,
(2) The *covering number* $\mathrm{cov}(\mathcal{I}_\kappa) = \min\big\{|\mathcal{C}| \;\big|\; \mathcal{C} \subseteq \mathcal{I}_\kappa \text{ and } \bigcup \mathcal{C} = {}^\kappa 2\big\}$,
(3) The *uniformity number* $\mathrm{non}(\mathcal{I}_\kappa) = \min\big\{|N| \;\big|\; N \subseteq {}^\kappa 2 \text{ and } N \notin \mathcal{I}_\kappa\big\}$,
(4) The *cofinality number* $\mathrm{cof}(\mathcal{I}_\kappa) = \min\big\{|\mathcal{F}| \;\big|\; \mathcal{F} \subseteq \mathcal{I}_\kappa \text{ and } \forall X \in \mathcal{I}_\kappa\, \exists Y \in \mathcal{F}\,(X \subseteq Y)\big\}$.

The following properties are well-known and easy to see. Note that the last two properties imply various relations between the uniformity and covering numbers of the ideals given in Figure 1, reflecting the relations displayed in that diagram.

Proposition 4.2. Let $\mathcal{I}_\kappa \subseteq \mathcal{J}_\kappa$ be $\leq\kappa$-complete ideals on ${}^\kappa 2$.

(1) $\kappa^+ \leq \mathrm{add}(\mathcal{I}_\kappa) \leq \min\{\mathrm{cov}(\mathcal{I}_\kappa), \mathrm{non}(\mathcal{I}_\kappa)\}$,
(2) $\max\{\mathrm{cov}(\mathcal{I}_\kappa), \mathrm{non}(\mathcal{I}_\kappa)\} \leq \min\{2^\kappa, \mathrm{cof}(\mathcal{I}_\kappa)\}$,
(3) $\mathrm{cof}(\mathcal{I}_\kappa) \leq 2^{2^\kappa}$,
(4) $\mathrm{non}(\mathcal{I}_\kappa) \leq \mathrm{non}(\mathcal{J}_\kappa)$,
(5) $\mathrm{cov}(\mathcal{J}_\kappa) \leq \mathrm{cov}(\mathcal{I}_\kappa)$. □

4.1. Comparison between the higher closed null cardinal characteristics.

We will first compare the cardinal characteristics of the families defined in Definition 2.7 to each other. The following theorem gives a general condition for comparing the cardinal characteristics of the ideal $\mathcal{E}_\kappa(B)$ with those of $\mathcal{E}_\kappa(C)$, for $B, C \in \mathrm{BP}_\kappa^{\lim}$.

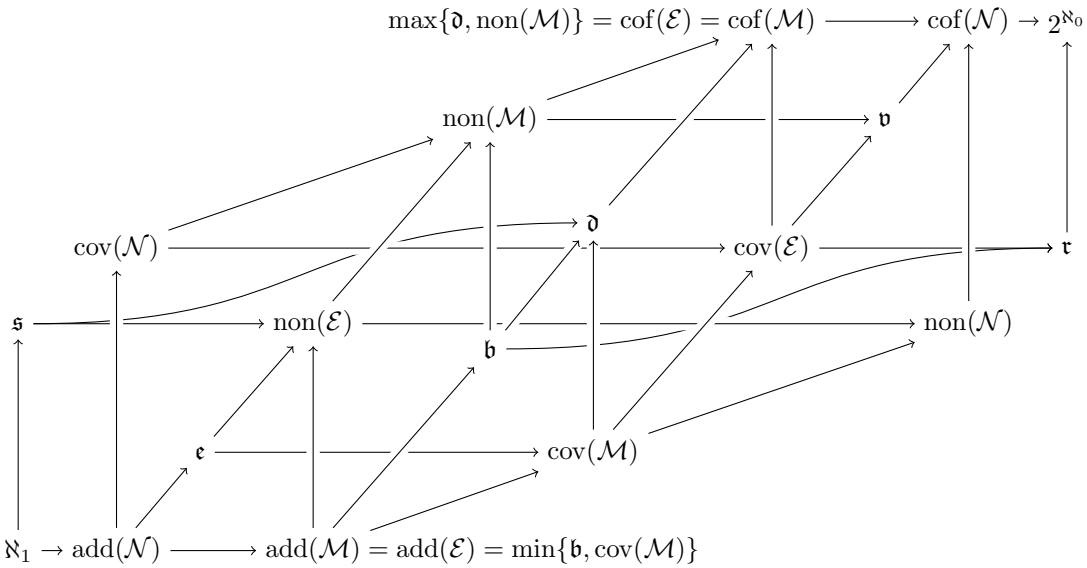


FIGURE 2. A diagram of ZFC-provable relations between cardinal characteristics of the (classical) continuum related to the ideal $\mathcal{E}$. An arrow $\mathfrak{x} \to \mathfrak{y}$ means that $\mathfrak{x} \leq \mathfrak{y}$ is provable.

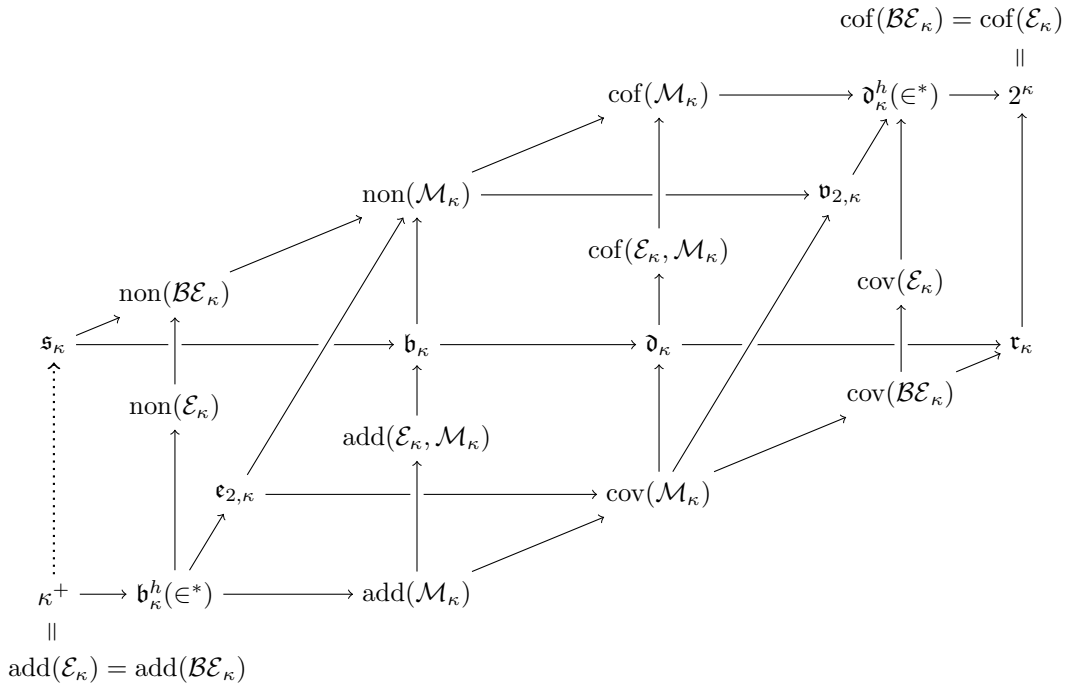


FIGURE 3. A diagram of ZFC-provable relations between cardinal characteristics of the higher continuum under the assumption that κ is strongly inaccessible. The dotted arrow requires the assumption that κ is weakly compact.

Theorem 4.3. If $B, C \in \mathrm{BP}^{\mathrm{lim}}_\kappa$ and $f \in {}^\kappa\kappa$ is injective such that $2^{|B_\alpha|} = 2^{|C_{f(\alpha)}|}$ for almost all $\alpha \in \kappa$, then

$$\mathrm{non}(\mathcal{E}_\kappa(C)) \leq \mathrm{non}(\mathcal{E}_\kappa(B)), \qquad \mathrm{cov}(\mathcal{E}_\kappa(B)) \leq \mathrm{cov}(\mathcal{E}_\kappa(C)),$$
$$\mathrm{add}(\mathcal{E}_\kappa(C)) \leq \mathrm{add}(\mathcal{E}_\kappa(B)), \qquad \mathrm{cof}(\mathcal{E}_\kappa(B)) \leq \mathrm{cof}(\mathcal{E}_\kappa(C)).$$

Proof. We first show that $\mathrm{non}(\mathcal{E}_\kappa(B)) \geq \mathrm{non}(\mathcal{E}_\kappa(C))$ and remark that $\mathrm{cov}(\mathcal{E}_\kappa(B)) \leq \mathrm{cov}(\mathcal{E}_\kappa(C))$ is proved by the same argument. Suppose $B, C \in \mathrm{BP}^{\lim}_\kappa$, $\alpha_0 \in \kappa$ and $f \in {}^\kappa\kappa$ are such that f is injective and $2^{|B_\alpha|} = 2^{|C_{f(\alpha)}|}$ for all $\alpha \in [\alpha_0, \kappa)$. For each $\alpha \in [\alpha_0, \kappa)$ we fix a bijection $\pi_\alpha \colon {}^{B_\alpha}2 \to {}^{C_{f(\alpha)}}2$.

Let us fix some $X \subseteq {}^\kappa 2$ with $X \notin \mathcal{E}_\kappa(B)$. For each $x \in X$, let $x' \in {}^\kappa 2$ be defined by

$$x' \colon \xi \mapsto \begin{cases} \pi_\alpha(x \restriction B_\alpha)(\xi) & \text{if there is } \alpha \in [\alpha_0, \kappa) \text{ such that } \xi \in C_{f(\alpha)} \\ 0 & \text{otherwise.} \end{cases}$$

Let $X' = \{x' \mid x \in X\}$, then $|X'| \leq |X|$. Let $\varphi' \in \Sigma_\kappa(C)$, then we define φ by

$$\varphi \colon \alpha \mapsto \begin{cases} \pi_\alpha^{-1}\big[\varphi'(f(\alpha))\big] & \text{if } \alpha \in [\alpha_0, \kappa) \\ \varnothing & \text{otherwise.} \end{cases}$$

Then $\varphi \in \Sigma_\kappa(B)$, since $|\varphi(\alpha)| \leq |\varphi'(f(\alpha))| < 2^{|C_{f(\alpha)}|} = 2^{|B_\alpha|}$. Because $X \notin \mathcal{E}_\kappa(B)$, there is $x \in X \setminus [\varphi]_*$, and thus $\exists^\infty \alpha \in \kappa \big(x \restriction B_\alpha \notin \varphi(\alpha)\big)$. However, note that $x' \restriction C_{f(\alpha)} = \pi_\alpha(x \restriction B_\alpha)$, therefore $\exists^\infty \alpha \big(x' \restriction C_\alpha \notin \varphi'(\alpha)\big)$, and thus $X' \not\subseteq [\varphi']_*$. Since φ' was arbitrary, we see that $X' \notin \mathcal{E}_\kappa(C)$.

Now for $\mathrm{add}(\mathcal{E}_\kappa(C)) \leq \mathrm{add}(\mathcal{E}_\kappa(B))$, given any $\psi \in \Sigma_\kappa(B)$ we define

$$\psi' \colon \xi \mapsto \begin{cases} \pi_\alpha\big[\psi(\alpha)\big] & \text{if } \xi \in \mathrm{ran}(f) \text{ and } \alpha = f^{-1}(\xi), \\ \varnothing & \text{otherwise,} \end{cases}$$

and given $\varphi' \in \Sigma_\kappa(C)$, we define $\varphi \in \Sigma_\kappa(B)$ as above. Then note that $\psi'(f(\alpha)) \subseteq \varphi'(f(\alpha))$ implies $\psi(\alpha) \subseteq \varphi(\alpha)$. Now, if $\psi_\beta \in \Sigma_\kappa(B)$ for each $\beta \in \gamma$, where $\gamma < \mathrm{add}(\mathcal{E}_\kappa(C))$, then $\bigcup_{\beta\in\gamma}\big[\psi'_\beta\big] \in \mathcal{E}_\kappa(C)$, and thus there is some $\varphi' \in \Sigma_\kappa(C)$ such that $\psi'_\beta(f(\alpha)) \subseteq \varphi'(f(\alpha))$ for almost all $\alpha \in \kappa$, from which we see that $\psi_\beta(\alpha) \subseteq \varphi(\alpha)$ for almost all α, and thus $\bigcup_{\beta\in\gamma}\big[\psi_\beta\big]_* \subseteq [\varphi]_* \in \mathcal{E}_\kappa(B)$. Therefore $\gamma < \mathrm{add}(\mathcal{E}_\kappa(B))$. □

Corollary 4.4. For successor κ and $B \in \mathrm{BP}^{\lim}_\kappa$, $\mathrm{non}(\mathcal{E}_\kappa(B)) = \mathrm{non}(\mathcal{E}_\kappa)$ and $\mathrm{cov}(\mathcal{E}_\kappa(B)) = \mathrm{cov}(\mathcal{E}_\kappa)$.

Proof. For κ successor, $B \in \mathrm{BP}^{\lim}_\kappa$ if and only if $|B_\alpha| = \lambda$ for almost all α. Moreover, by Corollary 2.25, $\mathcal{E}_\kappa = \mathcal{E}_\kappa(B)$ for some $B \in \mathrm{BP}^{\lim}_\kappa$. □

Theorem 4.5. For each $B \in \mathrm{BP}^{\lim}_\kappa$, $\mathrm{cof}(\mathcal{E}_\kappa(B)) \leq 2^\kappa$ and $\mathrm{cof}(\mathcal{E}_\kappa) \leq 2^\kappa$ and $\mathrm{cof}(\mathcal{BE}_\kappa) \leq 2^\kappa$.

Proof. Remember that we assume $\kappa = \kappa^{<\kappa}$. Now $\Sigma_\kappa(B) \subseteq \prod_{\alpha\in\kappa}\big[{}^{B_\alpha}2\big]^{<2^{|B_\alpha|}}$, and since $|B_\alpha| < \kappa$, we see that $\big|{}^{B_\alpha}2\big| \leq 2^{<\kappa} = \kappa$, and thus $\Big|\big[{}^{B_\alpha}2\big]^{<2^{|B_\alpha|}}\Big| \leq \kappa^{<\kappa} = \kappa$. That is, $|\Sigma_\kappa(B)| = 2^\kappa$. Moreover, $\big|\mathrm{BP}^{\lim}_\kappa\big| = 2^\kappa$, since $\mathrm{BP}^{\lim}_\kappa \subseteq \prod_{\alpha\in\kappa}[\kappa]^{<\kappa}$ and $\big|[\kappa]^{<\kappa}\big| = \kappa$. □

Theorem 4.6. There exist some $I \in \mathrm{IP}^{\lim}_\kappa$ and $B \in \mathrm{BP}^{\lim}_\kappa$ for which $\mathrm{add}(\mathcal{E}_\kappa(I)) \leq \mathrm{add}(\mathcal{E}^\forall_\kappa)$ and $\mathrm{add}(\mathcal{E}_\kappa(B)) \leq \mathrm{add}(\mathcal{BE}^\forall_\kappa)$. Moreover, there exist some $I \in \mathrm{IP}^{\lim}_\kappa$ and $B \in \mathrm{BP}^{\lim}_\kappa$ for which $\mathrm{non}(\mathcal{BE}^\forall_\kappa) = \mathrm{non}(\mathcal{E}_\kappa(B))$ and $\mathrm{non}(\mathcal{E}^\forall_\kappa) = \mathrm{non}(\mathcal{E}_\kappa(I))$.

Proof. By definition: suppose $\langle X_\alpha \mid \alpha \in \mathrm{add}(\mathcal{E}^\forall_\kappa)\rangle$ is a sequence in $\mathcal{E}^\forall_\kappa$ such that $\bigcup_\alpha X_\alpha \notin \mathcal{E}^\forall_\kappa$, then there is $I \in \mathrm{IP}^{\lim}_\kappa$ such that $\bigcup_\alpha X_\alpha \notin \mathcal{E}_\kappa(I)$. Similarly for the uniformity numbers. □

In terms of additivity numbers, Theorems 2.28 and 2.31 showed that $\mathrm{add}(\mathcal{BE}^\exists_\kappa) = 2$ and (under assumptions on κ) $\mathrm{add}(\mathcal{E}^\exists_\kappa) = 2$. We will use the same method of proof to determine the values of $\mathrm{add}(\mathcal{BE}_\kappa)$, $\mathrm{add}(\mathcal{E}_\kappa)$, $\mathrm{cof}(\mathcal{BE}_\kappa)$ and $\mathrm{cof}(\mathcal{E}_\kappa)$. Note that the higher case does not follow our classical intuition, where $\mathrm{add}(\mathcal{E}) = \mathrm{add}(\mathcal{M})$ and $\mathrm{cof}(\mathcal{E}) = \mathrm{cof}(\mathcal{M})$.

Theorem 4.7. Assume $\kappa = \kappa^{<\kappa}$.

(1) $\mathrm{add}(\mathcal{BE}_\kappa) = \kappa^+$.
(2) $\mathrm{cof}(\mathcal{BE}_\kappa) = 2^\kappa$.

(3) If the continuum function is not eventually constant below κ, then $\mathrm{add}(\mathcal{E}_\kappa) = \kappa^+$.
(4) If the continuum function is not eventually constant below κ, then $\mathrm{cof}(\mathcal{E}_\kappa) = 2^\kappa$.

Proof. (1). By recursion we may construct a sequence $\langle B^\xi \mid \xi \in \kappa^+\rangle$ such that for any distinct $\xi, \xi' \in \kappa^+$ and for almost all $\alpha, \beta \in \kappa$ we have $\left|B^\xi_\alpha \cap B^{\xi'}_\beta\right| \leq 1$. For each ξ we define $\varphi^\xi \in \Sigma_\kappa(B^\xi)$ such that for any $\alpha \in \kappa$, any $\eta \in B_\alpha$ and any $i \in \{0,1\}$, there is some $s \in \varphi^\xi(\alpha)$ with $s(\eta) = i$. Then we claim that $\bigcup_{\xi\in\kappa^+}[\varphi^\xi]_* \notin \mathcal{BE}_\kappa$.

Suppose, towards contradiction, that $\langle C^\eta \mid \eta \in \kappa\rangle$ is a sequence in $\mathrm{BP}^{\mathrm{lim}}_\kappa$ and $\langle \psi^\eta \mid \eta \in \kappa\rangle$ is such that $\psi^\eta \in \Sigma_\kappa(C^\eta)$ for each $\eta \in \kappa$ and $\bigcup_{\xi\in\kappa^+}[\varphi^\xi]_* \subseteq \bigcup_{\eta\in\kappa}[\psi^\eta]_*$. For each $\xi \in \kappa^+$, we see that $[\varphi^\xi]_0 = \bigcup_{\eta\in\kappa}\big([\psi^\eta]_* \cap [\varphi^\xi]_0\big)$ and, by Lemma 2.20, $\mathcal{M}_\kappa([\varphi^\xi]_0)$ is $\leq\kappa$-complete and proper, therefore for each $\xi \in \kappa^+$ there is $\eta_\xi \in \kappa$ for which $[\psi^{\eta_\xi}]_* \cap [\varphi^\xi]_0 \notin \mathcal{M}_\kappa([\varphi^\xi]_0)$. Then by the pigeonhole principle there are distinct $\xi, \xi' \in \kappa^+$ such that $\eta := \eta_\xi = \eta_{\xi'}$. But this contradicts Lemma 2.27, as B^ξ, $B^{\xi'}$, φ^ξ and $\varphi^{\xi'}$ satisfy all of its assumptions.

(2). The idea is dual to (1). We show that there exists a sequence $\langle B^\xi \mid \xi \in 2^\kappa\rangle$ in $\mathrm{BP}^{\mathrm{lim}}_\kappa$ such that for any distinct $\xi, \xi' \in \kappa^+$ and almost all $\alpha, \beta \in \kappa$ we have $\left|B^\xi_\alpha \cap B^{\xi'}_\beta\right| \leq 1$. This suffices: if we define φ^ξ for each ξ as above, and we fix a sequence $\langle C^\eta \mid \eta \in \lambda\rangle$ in $\mathrm{BP}^{\mathrm{lim}}_\kappa$ with $\lambda < 2^\kappa$, as well as corresponding $\psi^\eta \in \Sigma_\kappa(C^\eta)$, then Lemma 2.27 shows that for each η there is at most one $\xi \in 2^\kappa$ for which $[\psi^\eta]_* \cap [\varphi^\xi]_0 \notin \mathcal{M}_\kappa([\varphi^\xi]_0)$, and therefore, no κ-unions of sets of the form $[\psi^\eta]_*$ will be able to cover more than κ-many sets of the form $[\varphi^\xi]_*$, that is, $\mathrm{cof}(\mathcal{BE}_\kappa) > \lambda$.

To conclude, we argue that a sequence $\langle B^\xi \mid \xi \in 2^\kappa\rangle$ in $\mathrm{BP}^{\mathrm{lim}}_\kappa$ with the above properties exists. We fix an almost disjoint family $\{A_\xi \mid \xi \in 2^\kappa\} \subseteq [\kappa]^\kappa$, that is, a family such that $|A_\xi \cap A_{\xi'}| < \kappa$ for all distinct $\xi, \xi' \in 2^\kappa$. We also fix $C \in \mathrm{BP}^{\mathrm{lim}}_\kappa$. Now, for any $\xi \in 2^\kappa$, let $\langle a^\xi_\alpha \mid \alpha \in \kappa\rangle$ and $\langle b^\xi_\alpha \mid \alpha \in \kappa\rangle$ enumerate A_ξ and $\bigcup_{\alpha\in\kappa\setminus A_\xi} C_\alpha$, respectively, and define $B^\xi_\alpha = C_{a^\xi_\alpha} \cup \{b^\xi_\alpha\}$.

It is easy to see that for distinct $\xi, \eta \in 2^\kappa$ and for large enough $\alpha, \delta \in \kappa$ we have $\left|B^\xi_\alpha \cap B^\eta_\delta\right| \leq 2$. This is actually good enough to prove a corresponding version of Lemma 2.27, or alternatively we can bring the ≤ 2 down to ≤ 1 with a little more precision in the set-up. It suffices to assume that $C \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ (i.e., consists of consecutive intervals), that the enumerations $\langle a^\xi_\alpha \mid \alpha \in \kappa\rangle$ and $\langle b^\xi_\alpha \mid \alpha \in \kappa\rangle$ are increasing, and each A_ξ is sufficiently sparse to ensure $b^\xi_\alpha < \min\big(C_{a^\xi_\alpha}\big)$. Under these additional assumptions, for large enough α, δ we would have $\left|B^\xi_\alpha \cap B^\eta_\delta\right| = 2$ if and only if $B^\xi_\alpha \cap B^\eta_\delta = \{b^\xi_\alpha, b^\eta_\delta\}$, but this implies $b^\xi_\alpha \in C_{a^\eta_\delta}$ and $b^\eta_\delta \in C_{a^\xi_\alpha}$, contradicting that both $b^\xi_\alpha < \min\big(C_{a^\xi_\alpha}\big)$ and $b^\eta_\delta < \min\big(C_{a^\eta_\delta}\big)$.

(3) & (4). Similar to (1) & (2), but we use families $\langle x^\xi \mid \xi \in \kappa^+\rangle$ and $\langle x^\xi \mid \xi \in 2^\kappa\rangle$, respectively, of strictly increasing functions in ${}^\kappa\kappa$ whose ranges are mutually almost disjoint, and then apply Lemma 2.29 instead of Lemma 2.27. □

4.2. **Comparison with other cardinal characteristics.** As a starting ground, we could consider the cardinal characteristics of the classical closed null ideal, such as displayed in Figure 2. See [6, 3] for the classical definitions of the cardinal characteristics and proofs of the theorems involved. With the exception of $\mathrm{cov}(\mathcal{N})$ and $\mathrm{non}(\mathcal{N})$, each of the cardinal characteristics from Figure 2 has a natural generalisation to the higher context of ${}^\kappa 2$, and we will introduce these in the relevant sections. We will look at relations with the bounding and dominating numbers (Section 4.2.1), splitting and reaping numbers (Section 4.2.2), localisation numbers (Section 4.2.3), and evasion and prediction numbers (Section 4.2.4), and finally we give a relation between cardinal characteristics of the ideal $\mathcal{NA}_\kappa$ and the ideals $\mathcal{E}_\kappa(I)$ (Section 4.2.5).

4.2.1. *The bounding and dominating numbers.* We refer to [13, 8] for more information on the higher bounding and dominating numbers, here we just provide the definition and the known relations to the cardinal characteristics of $\mathcal{M}_\kappa$.

Definition 4.8. For $f, g \in {}^{\kappa}\kappa$, let $f \leq^* g$ if and only if $\big|\{\alpha \in \kappa \mid g(\alpha) < f(\alpha)\}\big| < \kappa$. We call a set $B \subseteq {}^{\kappa}\kappa$ *unbounded* if no $g \in {}^{\kappa}\kappa$ exists such that $f \leq^* g$ for all $f \in B$, and we call a set $D \subseteq {}^{\kappa}\kappa$ *dominating* if every $f \in {}^{\kappa}\kappa$ has some $g \in D$ such that $f \leq^* g$. The *bounding number* $\mathfrak{b}_\kappa$ is the least size of an unbounded subset of ${}^{\kappa}\kappa$ and the *dominating number* $\mathfrak{d}_\kappa$ is the least size of a dominating subset of ${}^{\kappa}\kappa$.

Fact 4.9 (Cummings and Shelah [13], Brendle et al. [8])**.** Assume $\kappa = \kappa^{<\kappa}$.q

(1) $\mathfrak{b}_\kappa \leq \mathfrak{d}_\kappa$,
(2) $\mathfrak{b}_\kappa \leq \mathrm{non}(\mathcal{M}_\kappa)$ and $\mathrm{cov}(\mathcal{M}_\kappa) \leq \mathfrak{d}_\kappa$,
(3) $\mathrm{add}(\mathcal{M}_\kappa) = \min\{\mathfrak{b}_\kappa, \mathrm{cov}(\mathcal{M}_\kappa)\}$ and $\mathrm{cof}(\mathcal{M}_\kappa) = \max\{\mathfrak{d}_\kappa, \mathrm{non}(\mathcal{M}_\kappa)\}$.

We will need the following generalisations of the additivity and cofinality numbers, which we will use to generalise a result by Miller [22] (see also [3, Theorems 2.6.10 and 2.2.2] for classical analogues).

Definition 4.10. Suppose that $\mathcal{J}$ is a subset of $\mathcal{I}$, then $\mathrm{add}(\mathcal{J},\mathcal{I})$ denotes the least size of a set $J \subseteq \mathcal{J}$ such that $\bigcup J \notin \mathcal{I}$, and $\mathrm{cof}(\mathcal{J},\mathcal{I})$ denotes the least size of a family $\mathcal{F} \subseteq \mathcal{I}$ such that every $J \in \mathcal{J}$ is contained in some $I \in \mathcal{F}$.

Fact 4.11. If $\mathcal{J} \subseteq \mathcal{K} \subseteq \mathcal{I}$, then $\mathrm{add}(\mathcal{K}) \leq \mathrm{add}(\mathcal{J},\mathcal{I})$ and $\mathrm{cof}(\mathcal{K}) \geq \mathrm{cof}(\mathcal{J},\mathcal{I})$. □

Theorem 4.12. Let κ be strongly inaccessible, then $\mathrm{add}(\mathcal{E}_\kappa, \mathcal{M}_\kappa) \leq \mathfrak{b}_\kappa$ and $\mathrm{cof}(\mathcal{E}_\kappa, \mathcal{M}_\kappa) \geq \mathfrak{d}_\kappa$.

Proof. We prove that $\mathrm{add}(\mathcal{E}_\kappa, \mathcal{M}_\kappa) \leq \mathfrak{b}_\kappa$, and remark that $\mathrm{cof}(\mathcal{E}_\kappa, \mathcal{M}_\kappa) \geq \mathfrak{d}_\kappa$ is proved similarly.

Let $B \subseteq {}^{\kappa}\kappa$ be an unbounded set. Fix a sequence of cardinals $\langle \kappa_\alpha \mid \alpha \in \kappa\rangle$ such that $\langle 2^{\kappa_\alpha} \mid \alpha \in \kappa\rangle$ is strictly increasing. Given $b \in B$ we define $b' \in {}^{\kappa}\kappa$ by $b'(0) = 0$, and for limit $\gamma \in \kappa$ let $b'(\gamma) = \sup_{\alpha\in\gamma} b'(\alpha)$, and finally

$$b'(\alpha+1) = \sup\big(\{b'(\alpha)+1\} \cup \{b(\xi) \mid \xi \leq b'(\alpha)\}\big). \tag{1}$$

We will write $\kappa^b_\alpha := \kappa_{b'(\alpha)}$ for all $\alpha \in \kappa$. Note that b' (and thus also $\langle \kappa^b_\alpha \mid \alpha \in \kappa\rangle$) is strictly increasing and continuous. Now we will proceed in a similar manner to the proof of Theorem 3.11. Define the sets

$$\begin{aligned} X^b &:= \big\{x \in {}^{\kappa}2 \;\big|\; \forall\alpha \in \kappa\, \forall\xi \in \big[\kappa^b_{2\alpha}, \kappa^b_{2\alpha+1}\big)\,(x(\xi) = 0)\big\},\\ Y^b &:= \big\{x \in {}^{\kappa}2 \;\big|\; \forall\alpha \in \kappa\, \forall\xi \in \big[\kappa^b_{2\alpha+1}, \kappa^b_{2\alpha+2}\big)\,(x(\xi) = 0)\big\},\\ Z &:= \big\{X^b \;\big|\; b \in B\big\} \cup \big\{Y^b \;\big|\; b \in B\big\}. \end{aligned}$$

We first show that $Z \subseteq \mathcal{E}^{\exists}_\kappa$. Namely, let us define

$$J_\alpha = [\kappa^b_{2\alpha}, \kappa^b_{2\alpha+2}), \qquad I_\alpha = \begin{cases} [\kappa^b_{2\alpha}, \kappa^b_{2\alpha+1}) & \text{if } \alpha \text{ is limit,}\\ [\kappa^b_{2\alpha+1}, \kappa^b_{2\alpha+3}) & \text{otherwise,}\end{cases}$$

then $I, J \in \mathrm{IP}^{\lim}_\kappa$ and $X^b \in \mathcal{E}_\kappa(I)$ and $Y^b \in \mathcal{E}_\kappa(J)$.

To finish the proof, it suffices to show that $\bigcup Z \notin \mathcal{M}_\kappa$. Suppose otherwise, and let $\langle N_\eta \mid \eta \in \kappa\rangle$ be a sequence of nowhere dense subsets of ${}^{\kappa}2$ such that $\bigcup Z = \bigcup_{\eta\in\kappa} N_\eta$. Using that κ is strongly inaccessible, we inductively build a continuous sequence of ordinals $\langle \xi_\eta \mid \eta \in \kappa\rangle$ and a sequence $\langle s_\eta \mid \eta \in \kappa\rangle$ of elements of ${}^{<\kappa}2$ such that for each $\eta \in \kappa$,

(1) if $t\colon \kappa_{\xi_\eta} \to 2$, then $[t^\frown s_\eta]$ is disjoint from $\bigcup_{\beta<\eta} N_\eta$ (note that this is possible, since $\bigcup_{\beta<\eta} N_\eta \in \mathcal{NWD}$, by $<\kappa$-completeness of $\mathcal{NWD}$), and
(2) $\kappa_{\xi_{\eta+1}}$ is larger than the ordinal sum $\kappa_{\xi_\eta} + \mathrm{dom}(s_\eta)$.

Define $g \in {}^{\kappa}\kappa$ by

$$g\colon \beta \mapsto \min\{\gamma \in \kappa \mid \exists \eta \in \kappa\, (\beta \leq \xi_\eta \text{ and } \xi_{\eta+2} \leq \gamma)\}.$$

Since B is unbounded, there is some $b \in B$ such that $b \not\leq^* g$, and thus the set $\{\beta \in \kappa \mid g(\beta) < b(\beta)\}$ has size κ. Fix an element β such that $g(\beta) < b(\beta)$, and let $\alpha \in \kappa$ be minimal such that $\beta \leq b'(\alpha)$, with b' defined from b as in (1), then by definition of b' we have $b(\beta) \leq b'(\alpha+1)$. Now by definition of g, there exists $\eta \in \kappa$ such that

$$\beta \leq \xi_\eta < \xi_{\eta+1} < \xi_{\eta+2} \leq g(\beta) < b(\beta) \leq b'(\alpha+1).$$

We consider two cases. Either $b'(\alpha) \leq \xi_{\eta+1} < \xi_{\eta+2} < b'(\alpha+1)$, or else $\xi_{\eta+1} < b'(\alpha)$ and thus by minimality of α we see that α is successor, say $\alpha = \alpha' + 1$ with $b(\alpha') < \beta$, and then we see that

$$b'(\alpha') < \beta \leq \xi_\eta < \xi_{\eta+1} < b'(\alpha) = b'(\alpha'+1).$$

In conclusion, for each β with $g(\beta) < b(\beta)$, there are $\alpha, \eta \in \kappa$ with $b'(\alpha) \leq \xi_\eta < \xi_{\eta+1} < b'(\alpha+1)$. Let us define the sets

$$A_0 := \{(\alpha,\eta) \in \kappa \times \kappa \mid b'(2\alpha) \leq \xi_\eta < \xi_{\eta+1} < b'(2\alpha+1)\},$$
$$A_1 := \{(\alpha,\eta) \in \kappa \times \kappa \mid b'(2\alpha+1) \leq \xi_\eta < \xi_{\eta+1} < b'(2\alpha+2)\},$$

then $|A_0 \cup A_1| = \kappa$, thus let us assume without loss that $|A_0| = \kappa$. We define $x \in {}^{\kappa}2$ by

$$x(\zeta) = \begin{cases} s_\eta(\zeta') & \text{if there are } (\alpha,\eta) \in A_0 \text{ and } \zeta' \in \mathrm{dom}(s_\eta) \text{ with } \zeta = \kappa_{\xi_\eta} + \zeta', \\ 0 & \text{otherwise,} \end{cases}$$

then x is only nonzero in intervals $[\kappa_{\xi_\eta}, \kappa_{\xi_{\eta+1}})$ for which there exists an $\alpha \in \kappa$ with $(\alpha,\eta) \in A_0$. Therefore x is only nonzero on intervals $\left[\kappa^b_{2\alpha}, \kappa^b_{2\alpha+1}\right)$, that is, $x \in Y^b \subseteq \bigcup Z$. But simultaneously, for each $(\alpha,\eta) \in A_0$ we have $[x \restriction (\kappa_{\xi_\eta} + \mathrm{dom}(s_\eta))] = [(x \restriction \kappa_{\xi_\eta})^\frown s_\eta]$, which is disjoint from $\bigcup_{\beta<\eta} N_\eta$ by our definitions of ξ_η and s_η. But then $x \notin \bigcup_{\eta\in\kappa} N_\eta = \bigcup Z$, which is a contradiction. □

Although we already know the values of $\mathrm{add}(\mathcal{E}_\kappa)$, $\mathrm{add}(\mathcal{BE}_\kappa)$, $\mathrm{cof}(\mathcal{E}_\kappa)$ and $\mathrm{cof}(\mathcal{BE}_\kappa)$ from Theorem 4.7, Theorem 4.12 gives us new information, since we note that $\mathrm{add}(\mathcal{M}_\kappa) \leq \mathrm{add}(\mathcal{I}, \mathcal{M}_\kappa)$ for any $\mathcal{I} \subseteq \mathcal{M}_\kappa$, and thus consistently $\mathrm{add}(\mathcal{E}_\kappa, \mathcal{M}_\kappa) \neq \mathrm{add}(\mathcal{E}_\kappa)$, and so on.

4.2.2. *The splitting and reaping numbers.* The splitting number has been well-studied in the higher context, and behaves differently than our classical intuition would suggest.

Definition 4.13. For $X, Y \in [\kappa]^\kappa$, say that X splits Y if $|Y \cap X| = |Y \setminus X| = \kappa$. We call a family $S \subseteq [\kappa]^\kappa$ a *splitting family* if every $Y \in [\kappa]^\kappa$ is split by some $X \in S$, and we call a family $R \subseteq [\kappa]^\kappa$ an *unreapable family* if no $X \in [\kappa]^\kappa$ splits all $Y \in R$. The *splitting number* $\mathfrak{s}_\kappa$ is the least size of a splitting family, and the *reaping number* $\mathfrak{r}_\kappa$ is the least size of an unreapable family.

Fact 4.14 (Raghavan and Shelah [24], Suzuki [30], Zapletal [34])**.**

(1) $\mathfrak{s}_\kappa \leq \mathfrak{b}_\kappa \leq \mathfrak{r}_\kappa$ and if $\kappa \geq \beth_\omega$, then $\mathfrak{r}_\kappa \geq \mathfrak{d}_\kappa$,
(2) $\mathfrak{s}_\kappa \leq \mathrm{non}(\mathcal{M}_\kappa)$ and $\mathfrak{r}_\kappa \geq \mathrm{cov}(\mathcal{M}_\kappa)$,
(3) κ is strongly inaccessible if and only if $\kappa \leq \mathfrak{s}_\kappa$
(4) κ weakly compact if and only if $\kappa^+ \leq \mathfrak{s}_\kappa$,
(5) if $\kappa^+ < \mathfrak{s}_\kappa$, then there is an inner model with a measurable λ of Mitchell order $o(\lambda) \geq \lambda^{++}$.

It follows from this that, as a lower bound, the only cases in which it is interesting to consider $\mathfrak{s}_\kappa$ are when κ is at least weakly compact.

Theorem 4.15. We have $\mathfrak{s}_\kappa \leq \mathrm{non}(\mathcal{BE}_\kappa)$ and $\mathfrak{r}_\kappa \geq \mathrm{cov}(\mathcal{BE}_\kappa)$.

Proof. We prove that $\mathfrak{s}_\kappa \leq \mathrm{non}(\mathcal{BE}_\kappa)$ and remark that $\mathfrak{r}_\kappa \geq \mathrm{cov}(\mathcal{BE}_\kappa)$ is shown similarly.

Suppose that $X \subseteq {}^\kappa 2$ and $X \notin \mathcal{BE}_\kappa$ and let $X' \subseteq [\kappa]^{\leq\kappa}$ be defined as $X' = \{x^{-1}(1) \mid x \in X\}$; that is, X is the set of characteristic functions for the sets in X'. We claim that X' is a splitting family. Given $Y \subseteq [\kappa]^\kappa$ for which $|Y \setminus \kappa| = \kappa$, we choose some $B \in \mathrm{BP}_\kappa^{\lim}$ such that $|B_\alpha \cap Y| = 2^{|\alpha|}$ and $|B_\alpha \setminus Y| = |\alpha|$ for each $\alpha \in \kappa$, and we define $\varphi \in \Sigma_\kappa(B)$ such that

$$\varphi(\alpha) = \left\{ s \in {}^{B_\alpha}2 \;\middle|\; \forall \xi, \xi' \in B_\alpha \cap Y \, (s(\xi) = s(\xi')) \right\}.$$

Since $X \notin \mathcal{BE}_\kappa$, we may choose $x \in [X] \setminus [\varphi]_*$, then for κ-many α there are some $\xi_0, \xi_1 \in B_\alpha \cap Y$ such that $x(\xi_0) = 0$ and $x(\xi_1) = 1$. It follows that Y is split by $x^{-1}(1)$. □

We note that the above proof is highly specific to $\mathcal{BE}_\kappa$, and does not carry over to $\mathcal{E}_\kappa$.

4.2.3. *Localisation numbers.* We first introduce some terminology in order to define the localisation numbers. Consider a cardinal-valued function b with domain κ such that $b(\alpha) \geq 2$ for all $\alpha \in \kappa$, and let $\prod b$ denote the product $\prod_{\alpha\in\kappa} b(\alpha)$. If h is another cardinal-valued function with domain κ, then the elements of $\prod_{\alpha\in\kappa}[b(\alpha)]^{<h(\alpha)}$ are called (b,h)-*slaloms*. In the special case where $b = \overline{\kappa}$ is the constant function sending $\alpha \mapsto \kappa$ for all $\alpha \in \kappa$, we call a $(\overline{\kappa}, h)$-slalom simply an h-*slalom*.

For a (b,h)-slalom φ and $f \in \prod b$, we say that f is *localised* by φ if $\forall^\infty \alpha \in \kappa \, (f(\alpha) \in \varphi(\alpha))$, denoted by $f \in^* \varphi$. The (b,h)-*localisation number* $\mathfrak{d}_\kappa^{b,h}(\in^*)$ is the least cardinality of a family $\mathcal{F}$ of (b,h)-slaloms such that every $f \in \prod b$ is localised by a (b,h)-slalom in $\mathcal{F}$. Dually, the (b,h)-*avoidance number* $\mathfrak{b}_\kappa^{b,h}(\in^*)$ is the least cardinality of a family $\mathcal{A} \subseteq \prod b$ such that no single (b,h)-slalom localises all $f \in \mathcal{A}$. We write $\mathfrak{d}_\kappa^h(\in^*)$ and $\mathfrak{b}_\kappa^h(\in^*)$ if $b = \overline{\kappa}$.

Classically, localisation numbers were introduced by Bartoszyński [2] to provide a combinatorial characterisation of the cofinality and additivity numbers of the Lebesgue null ideal:

Fact 4.16 (Bartoszyński [2])**.** Let $h \in {}^\omega\omega$ be strictly increasing, then $\mathrm{add}(\mathcal{N}) = \mathfrak{b}_\omega^h(\in^*)$ and $\mathrm{cof}(\mathcal{N}) = \mathfrak{d}_\omega^h(\in^*)$.

The choice of parameters (b,h) is relevant classically, as Goldstern and Shelah [17], Kellner [21] showed that there can consistently be continuum many localisation numbers with distinct values. In the higher context, even if we fix $b = \overline{\kappa}$, the choice of h remains relevant, as shown in [8, Theorem 70]. In the higher context, localisation numbers were furthermore studied in [31, 5, 25].

The covering number of the ideal $\mathcal{E}_\kappa(B)$ can be expressed in terms of localisation numbers.

Theorem 4.17. Let $B \in \mathrm{BP}_\kappa$ and let $b \in {}^\kappa\kappa$ be defined by $b\colon \alpha \mapsto 2^{|B_\alpha|}$, then $\mathrm{cov}(\mathcal{E}_\kappa(B)) = \mathfrak{d}_\kappa^{b,b}(\in^*)$ and $\mathrm{non}(\mathcal{E}_\kappa(B)) = \mathfrak{b}_\kappa^{b,b}(\in^*)$.

Proof. This is not much more than a notational shuffle. Let B and b be as in the statement of the theorem, then we may consider the bijection $\pi\colon {}^\kappa 2 \to \prod b$ given by $x \mapsto \langle \pi_\alpha(x \restriction B_\alpha) \mid \alpha \in \kappa \rangle$, where each π_α is a bijection between ${}^{B_\alpha}2$ and $2^{|B_\alpha|}$. Similarly, if φ is a (b,b)-slalom, we define the slalom $\tilde{\pi}(\varphi)\colon \alpha \mapsto \pi_\alpha^{-1}[\varphi(\alpha)]$. The conclusion follows by noting that for any $x \in {}^\kappa 2$ and any (b,b)-slalom φ we have $\pi(x) \in^* \varphi$ if and only if $x \in [\tilde{\pi}(\varphi)]_*$. □

Corollary 4.18. If $\kappa = 2^{<\kappa}$ is successor, then $\mathrm{cov}(\mathcal{E}_\kappa) = \mathfrak{d}_\kappa$ and $\mathrm{non}(\mathcal{E}_\kappa) = \mathfrak{b}_\kappa$.

Proof. By Corollary 4.4 and Theorem 4.17 we have $\mathrm{cov}(\mathcal{E}_\kappa) = \mathrm{cov}(\mathcal{E}_\kappa(B)) = \mathfrak{d}_\kappa^{b,b}(\in^*)$ for some $B \in \mathrm{BP}_\kappa^{\lim}$ and suitably defined b. Since $2^{|B_\alpha|} = \kappa$ for large enough α and κ is regular, it is easy to see that $\mathfrak{d}_\kappa^{b,b}(\in^*) = \mathfrak{d}_\kappa$. □

In light of Fact 4.16, the following could be seen as a generalisation of the classical facts that $\mathrm{add}(\mathcal{N}) \leq \mathrm{non}(\mathcal{E})$ and $\mathrm{cov}(\mathcal{E}) \leq \mathrm{cof}(\mathcal{N})$.[3]

Theorem 4.19. Suppose $\kappa = 2^{<\kappa}$ is regular. Let h be cofinally increasing, then $\mathrm{cov}(\mathcal{E}_\kappa) \leq \mathfrak{d}^h_\kappa(\in^*)$ and $\mathfrak{b}^h_\kappa(\in^*) \leq \mathrm{non}(\mathcal{E}_\kappa)$.

Proof. Let h be cofinally increasing, let $I \in \mathrm{IP}^{\lim}_\kappa$ be such that $|I_\alpha| = h(\alpha)$, and let $b\colon \alpha \mapsto 2^{h(\alpha)}$. Then we have $\mathrm{cov}(\mathcal{E}_\kappa) \leq \mathrm{cov}(\mathcal{E}_\kappa(I)) = \mathfrak{d}^{b,b}_\kappa(\in^*) \leq \mathfrak{d}^{\overline{\kappa},b}_\kappa(\in^*) \leq \mathfrak{d}^h_\kappa(\in^*)$. □

4.2.4. *The evasion and prediction numbers.* Since we work on the higher Cantor space ${}^\kappa 2$, we will only consider binary evasion and prediction. Suppose we have a function $p : {}^{<\kappa}2 \to 2$ and $D \in [\kappa]^\kappa$, then $f \in {}^\kappa 2$ is *predicted* by p on D if for almost all $\alpha \in D$ we have $p(f \restriction \alpha) = f(\alpha)$, and otherwise f is *evaded* by p on D. We call the pair (p, D) a *predictor*. The (binary) *prediction number* $\mathfrak{v}_{2,\kappa}$ is the least number of predictors necessary to predict every $f \in {}^\kappa 2$, and the (binary) *evasion number* $\mathfrak{e}_{2,\kappa}$ is the least number of functions in ${}^\kappa 2$ such that every predictor is evaded by one of them.

It is fairly simple to see that the classical binary evasion number $\mathfrak{e}_2$ is at most of size $\mathrm{non}(\mathcal{E})$, as was first mentioned in [9, § 3.1]. Namely, if we fix a predictor (p, D) on ${}^\omega 2$, then the set $\{x \in {}^\omega 2 \mid \forall n \in D \setminus n_0\, (p(x \restriction n) = x(n))\}$ is easily seen to be closed and measure zero. On κ the situation is different, and our classical intuiton does not apply.

Theorem 4.20. There exists a predictor $(p, D) \in {}^{(<\kappa 2)}2 \times [\kappa]^\kappa$ such that the set of $f \in {}^\kappa 2$ predicted by (p, D) is not in $\mathcal{BE}_\kappa$.

Proof. Let $E = \{2 \cdot \alpha \mid \alpha \in \kappa\}$ be the set of even ordinals below κ and define $p : {}^{<\kappa}2 \to 2$ such that for any $\alpha \in E$ and $s \in {}^{\alpha+1}2$ we have $p(s) = s(\alpha)$. Then $(p, \kappa \setminus E)$ is such a predictor as claimed in the theorem statement. The reasoning in the proof of Theorem 2.10, which shows that $X \notin \mathcal{E}_\kappa$, applies in our situation to show that the set of $f \in {}^\kappa 2$ predicted by $(p, \kappa \setminus E)$ is not in $\mathcal{BE}_\kappa$. □

This suggests that the evasion number $\mathfrak{e}_{2,\kappa}$ may be unrelated to $\mathrm{non}(\mathcal{E}_\kappa)$.

4.2.5. *Uniformity and cofinality of the null additive ideal.* In this final subsection we will provide a relation between the uniformity number of $\mathcal{NA}_\kappa$ and the additivity number of the ideal $\mathcal{E}_\kappa(I)$ for $I \in \mathrm{IP}^{\lim}_\kappa$. A dual result shows a connection between the covering number of $\mathcal{NA}_\kappa$ and the cofinality number of $\mathcal{E}_\kappa(I)$. We can use this result to prove that $\mathrm{non}(\mathcal{NA}_\kappa) \leq \min\{\mathrm{add}(\mathcal{E}^\forall_\kappa), \mathrm{add}(\mathcal{BE}^\forall_\kappa)\}$ for strongly inaccessible κ.

Lemma 4.21. The following are equivalent for $X \subseteq {}^\kappa 2$:

(1) $X \in \mathcal{NA}_\kappa$.
(2) For each $f \in {}^\kappa\kappa$ with $\lim_{\alpha\in\kappa} f(\alpha) = \kappa$ and every $B \in \mathrm{BP}^{\lim}_\kappa$, there exists $\psi \in \prod_{\alpha\in\kappa} \left[{}^{B_\alpha}2\right]^{\leq |f(\alpha)|}$ such that $X \subseteq [\psi]_*$.

Proof. (2) ⇒ (1). This is clear by Definition 3.3, using $f : \alpha \mapsto |\alpha|$.

(1) ⇒ (2). Given $f \in {}^\kappa\kappa$ with $\lim_{\alpha\in\kappa} f(\alpha) = \kappa$ and $B \in \mathrm{BP}^{\lim}_\kappa$, let $\alpha_0 = 0$ and, for $\xi > 0$, let α_ξ be minimal such that

(a) $f(\alpha) \geq |\xi|$ for all $\alpha \geq \alpha_\xi$,
(b) $\alpha_\xi > \alpha_\eta$ for all $\eta \in \xi$, and
(c) $\bigcup_{\alpha\in[\alpha_\xi,\, \alpha_{\xi+1})} B_\alpha$ is an interval (which is possible by Lemma 2.13).

[3] Classically, these are corollaries of the fact that $\mathrm{add}(\mathcal{N}) \leq \mathrm{add}(\mathcal{M})$ and $\mathrm{cof}(\mathcal{M}) \leq \mathrm{cof}(\mathcal{N})$, but we remark that this is not immediately apparent in the higher context: although $\mathfrak{b}^h_\kappa(\in^*) \leq \mathrm{add}(\mathcal{M}_\kappa)$ and $\mathrm{cof}(\mathcal{M}_\kappa) \leq \mathfrak{d}^h_\kappa(\in^*)$ are provable for inaccessible κ, the situation under the weaker assumption that $\kappa = \kappa^{<\kappa}$ is unclear (see [8, Question 24]).

Since $\langle \alpha_\xi \mid \xi \in \kappa \rangle$ is continuous and strictly increasing, if we define $I_\xi := \bigcup_{\alpha \in [\alpha_\xi, \alpha_{\xi+1})} B_\alpha$, then $I \in \mathrm{IP}_\kappa^{\lim}$. Now let $\varphi \in \prod_{\xi \in \kappa} \left[{}^{I_\xi}2\right]^{|\xi|}$ be such that $X \subseteq [\varphi]_*$ and define $\psi(\alpha) = \{s \restriction B_\alpha \mid s \in \varphi(\xi)\}$ for each $\alpha \in [\alpha_\xi, \alpha_{\xi+1})$, then it is clear that $\psi \in \prod_{\alpha \in \kappa} \left[{}^{B_\alpha}2\right]^{\leq f(\alpha)}$ and $X \subseteq [\psi]_*$. □

Theorem 4.22. Let κ be strongly inaccessible and $B \in \mathrm{BP}_\kappa^{\lim}$, then $\mathrm{non}(\mathcal{NA}_\kappa) \leq \mathrm{add}(\mathcal{E}_\kappa(B))$ and $\mathrm{cof}(\mathcal{E}_\kappa(B)) \leq \mathrm{cov}(\mathcal{NA}_\kappa)$.

Proof. We will show that $\mathrm{non}(\mathcal{NA}_\kappa) \leq \mathrm{add}(\mathcal{E}_\kappa(B))$ and remark that $\mathrm{cof}(\mathcal{E}_\kappa(B)) \leq \mathrm{cov}(\mathcal{NA}_\kappa)$ is proved similarly.

Suppose $A \subseteq \mathcal{E}_\kappa(B)$ is a family of size $|A| < \mathrm{non}(\mathcal{NA}_\kappa)$, then we show that $\bigcup A \in \mathcal{E}_\kappa(B)$. We first pick for each $X \in A$ some $\varphi_X \in \Sigma_\kappa(B)$ such that $X \subseteq [\varphi_X]_*$. Given a set $C \in [\kappa]^{<\kappa}$, increasingly enumerated as $\langle c_\xi \mid \xi \in \gamma \rangle$, and a function $t \colon C \to 2$, let us write $\tilde{t} \colon \gamma \to 2$ defined by $\tilde{t} \colon \xi \mapsto t(c_\xi)$.

First, we encode each φ_X into a single element $x_X \in {}^\kappa 2$, then $\{x_X \mid X \in A\} \in \mathcal{NA}_\kappa$ by the assumption on $|A|$. For a fixed φ_X, pick an enumeration $\left\langle t_{X,\alpha,\xi} \;\middle|\; \xi \in |\varphi_X(\alpha)| \right\rangle$ of $\varphi_X(\alpha)$ for each $\alpha \in \kappa$, and define

$$s_{X,\alpha} := \tilde{t}_{X,\alpha,0}{}^\frown \tilde{t}_{X,\alpha,1}{}^\frown \cdots {}^\frown \tilde{t}_{X,\alpha,\xi}{}^\frown \cdots {}^\frown 1_\alpha,$$

where 1_α denotes the constant function with domain (the cardinal) $2^{|B_\alpha|}$ and value 1. We note that also $s_{X,\alpha}$ has domain $2^{|B_\alpha|}$. We define $x_X \in {}^\kappa 2$ by

$$x_X := s_{X,0}{}^\frown s_{X,1} \cdots {}^\frown s_{X,\alpha}{}^\frown \cdots .$$

Now define $J \in \mathrm{IP}_\kappa^{\lim}$ to be the (unique) interval partition such that each J_α is an interval whose order-type is the cardinal $2^{|B_\alpha|}$. We also (uniquely) partition each J_α into $\langle J_{\alpha,\beta} \mid \beta \in 2^{|B_\alpha|} \rangle$, where each $J_{\alpha,\beta}$ is an interval of order-type $|B_\alpha|$ and $\min(J_{\alpha,\beta}) < \min(J_{\alpha,\beta'})$ for all $\beta < \beta'$. By Lemma 4.21 and $|A| < \mathrm{non}(\mathcal{NA}_\kappa)$, we find $\psi \in \prod_{\alpha \in \kappa} \left[{}^{J_\alpha}2\right]^{\leq |B_\alpha|}$ such that $\{x_X \mid X \in A\} \subseteq [\psi]_*$.

Finally, we convert ψ into a slalom $\chi \in \Sigma_\kappa(B)$ such that $[\varphi_X]_* \subseteq [\chi]_*$ for each $X \in A$, by letting

$$\chi(\alpha) := \bigcup \left\{ t \in {}^{B_\alpha}2 \mid \exists s \in \psi(\alpha)\, \exists \beta < 2^{|B_\alpha|} \left(s \text{ is eventually constant and } \tilde{t} = (s \restriction J_{\alpha,\beta})^\sim \right) \right\}.$$

It is clear that $[\varphi_X]_* \subseteq [\chi]_*$. To see that $\chi \in \Sigma_\kappa(B)$, note that $|\psi(\alpha)| \leq |B_\alpha| < \mathrm{cf}(2^{|B_\alpha|})$, and that, if $s \in \psi(\alpha)$ is eventually constant, then $\left| \left\{ (s \restriction J_{\alpha,\beta})^\sim \;\middle|\; \beta \in 2^{|B_\alpha|} \right\} \right| < 2^{|B_\alpha|}$. □

Corollary 4.23. We have $\mathrm{non}(\mathcal{NA}_\kappa) \leq \mathrm{add}(\mathcal{E}_\kappa^\forall)$ and $\mathrm{non}(\mathcal{NA}_\kappa) \leq \mathrm{add}(\mathcal{BE}_\kappa^\forall)$. □

5. Consistency

In this section we will provide a few consistency results of strict inequalities between the cardinal characteristics of $\mathcal{E}_\kappa$ and $\mathcal{BE}_\kappa$ and some of the other cardinal characteristics from Figure 3. We will assume that κ is strongly inaccessible throughout this section.

There are many unknowns to consistency in the higher Baire space, as the theory of iterated forcing suffers from several complications, compared to the classical context. Two of the main problems are the absence of a suitable form of random forcing, and the difficulty of preserving properties like ${}^\kappa\kappa$-bounding under iteration. These two problems form two of the reasons that it is hard to find models in which $\mathrm{cov}(\mathcal{M}_\kappa) < \mathrm{non}(\mathcal{M}_\kappa)$.

The reverse strict inequality $\mathrm{non}(\mathcal{M}_\kappa) < \mathrm{cov}(\mathcal{M}_\kappa)$ is, on the other hand, easily obtained with κ-Cohen forcing, in exactly the same manner as is done classically. Moreover, κ-Hechler (dominating) forcing can be iterated with $<\kappa$-support to provide the consistency of $\mathfrak{b}_\kappa^h(\in^*) < \mathrm{add}(\mathcal{M}_\kappa)$ and of $\mathrm{cof}(\mathcal{M}_\kappa) < \mathfrak{d}_\kappa^h(\in^*)$. Finally, κ-localisation forcing proves the consistency of $\kappa^+ < \mathfrak{b}_\kappa^h(\in^*)$ and of $\mathfrak{d}_\kappa^h(\in^*) < 2^\kappa$. We refer to [8] for details about each of these constructions. As a corollary:

Theorem 5.1. Each of the inequalites $\kappa^+ < \mathrm{non}(\mathcal{E}_\kappa)$, $\mathrm{non}(\mathcal{BE}_\kappa) < \mathrm{cov}(\mathcal{M}_\kappa)$, $\mathrm{non}(\mathcal{M}_\kappa) < \mathrm{cov}(\mathcal{BE}_\kappa)$ and $\mathrm{cov}(\mathcal{E}_\kappa) < 2^\kappa$ is consistent for κ strongly inaccessible.

We recall the following properties of forcing notions.

(1) A forcing notion $\mathbb{P}$ is $<\kappa$-*distributive*, if $\mathbb{P}$ does not add any new functions $f : \kappa \to \mathbf{Ord}$.
(2) A forcing notion $\mathbb{P}$ is $<\kappa$-*closed*, if any $<\kappa$-length descending sequence of conditions has a lower bound.
(3) A forcing notion $\mathbb{P}$ is *strategically* $<\kappa$-*closed*, if you have a winning strategy for every $\lambda < \kappa$ in the following game. You and your opponent play a game of length λ, where each round $\alpha \in \lambda$ consists of a condition p^{Y}_α played by you, followed by a condition p^{O}_α played by your opponent. You must play $p^{\mathrm{Y}}_0 = \mathbb{1}$; is round α, you must play $p^{\mathrm{Y}}_\alpha \leq p^{\mathrm{O}}_\xi$ for all $\xi < \alpha$, and your opponent must play $p^{\mathrm{O}}_\alpha \leq p^{\mathrm{Y}}_\alpha$. You win if you do not run out of moves.
(4) A forcing notion $\mathbb{P}$ is $<\kappa^+$-*c.c.* if every antichain in $\mathbb{P}$ has cardinality at most λ.
(5) A subset $Q \subseteq \mathbb{P}$ is $<\lambda$-*centred* if every $P \subseteq Q$ with $|P| < \lambda$ has some $q \in \mathbb{P}$ with $q \leq p$ for all $p \in P$. A forcing notion $\mathbb{P}$ is $(\kappa, <\lambda)$-*centred* if there are $<\lambda$-centred sets $P_\alpha \subseteq \mathbb{P}$ for each $\alpha \in \kappa$ such that $\bigcup_{\alpha\in\kappa} P_\alpha$ is dense in $\mathbb{P}$.
(6) A subset $P \subseteq \mathbb{P}$ is λ-*calibre* if for every $P' \in [P]^\lambda$ there exists $q \in \mathbb{P}$ such that $\{p \in P' \mid q \leq p\}$ has size λ. A forcing notion $\mathbb{P}$ is (κ, λ)-*calibre* if there are λ-calibre sets $P_\alpha \subseteq \mathbb{P}$ for each $\alpha \in \kappa$ such that $\bigcup_{\alpha\in\kappa} P_\alpha$ is dense in $\mathbb{P}$.

Note that $<\kappa$-closed forcing notions are strategically so, and strategically $<\kappa$-closed forcing notions are $<\kappa$-distributive. Each of these three properties is preserved under $<\kappa$- and $\leq\kappa$-support iteration. If $\lambda \geq 3$, then $(\kappa, <\lambda)$-centred forcing notions are $<\kappa^+$-c.c. Although $<\kappa^+$-c.c. is not generally preserved under iterations, the combination of $(\kappa, <\lambda)$-centredness with $<\kappa$-closure and a canonicity assumption is preserved under $<\kappa$-support iteration of length $<(2^\kappa)^+$, see [8, Lemma 55]. Finally, $<\kappa$-distributive $<\kappa^+$-c.c. forcing notions preserve all cofinalities, and hence all cardinal numbers.

5.1. **Generic slaloms with localisation forcing.** In [8, Definition 50], κ-localisation forcing is defined. This forcing notion adds a generic slalom $\varphi \in \prod_{\alpha\in\kappa}[\kappa]^{|\alpha|}$ that localises all ground model functions $f \in {}^\kappa\kappa \cap \mathbf{V}$, in the sense that $f(\alpha) \in \varphi(\alpha)$ for almost all $\alpha \in \kappa$. We present two variations on this forcing notion.

5.1.1. *Adding a new slalom through an old partition.* The first forcing notion $\mathbb{LOC}_\kappa(B)$ adds a new slalom $\varphi \in \prod_{\alpha\in\kappa}\left[{}^{B_\alpha}2\right]^{|\alpha|}$ for a chosen ground model partition $B \in \mathrm{BP}^{\lim}_\kappa$.

Definition 5.2. Fix $B \in \mathrm{BP}^{\lim}_\kappa$. Every condition of $\mathbb{LOC}_\kappa(B)$ is a pair $p = (\varphi_p, \gamma_p)$ such that $\varphi_p \in \prod_{\alpha\in\kappa}\left[{}^{B_\alpha}2\right]^{|\alpha|}$ and $\gamma_p \in \kappa$. If $p, q \in \mathbb{LOC}_\kappa(B)$, then we define the ordering $q \leq p$ if and only if $\varphi_q \restriction \gamma_p = \varphi_p \restriction \gamma_p$ and $\gamma_q \geq \gamma_p$ and for each $\alpha \in \kappa \setminus \gamma_p$ we have $\varphi_q(\alpha) \supseteq \varphi_p(\alpha)$.

Lemma 5.3. If $B \in \mathrm{BP}^{\lim}_\kappa$, then $\mathbb{LOC}_\kappa(B)$ is strategically $<\kappa$-closed and $(\kappa, <\lambda)$-centred for $\lambda < \kappa$.

Proof. For each $\alpha \in \kappa$, let γ_α be minimal such that $\gamma_\alpha \geq |\alpha|$ and $|B_\xi| \geq \alpha$ for all $\xi \geq \gamma_\alpha$. For strategic closure, it suffices to note that $\{p \in \mathbb{LOC}_\kappa(B) \mid \gamma \geq \gamma_\lambda\}$ is $<\lambda$-closed. For centredness, if $\varphi \in \Sigma_\kappa(B)$ and $s = \varphi \restriction \gamma_\lambda$, then $\{p \in \mathbb{LOC}_\kappa(B) \mid \varphi_p \restriction \gamma_\lambda = s\}$ is $<\lambda$-centred. □

Let $\mathbf{V}$ be a model of ZFC containing $B \in \mathrm{BP}^{\lim}_\kappa$ and let G be $\mathbb{LOC}_\kappa(B)$-generic over $\mathbf{V}$, then we may define $\varphi_{\mathrm{gen}} := \bigcup_{p\in G}(\varphi_p \restriction \gamma_p)$ and note that $\varphi_{\mathrm{gen}} \in \prod_{\alpha\in\kappa}\left[{}^{B_\alpha}2\right]^{|\alpha|}$. Moreover, by genericity it is easy to see that $({}^\kappa 2)^{\mathbf{V}} \subseteq [\varphi_{\mathrm{gen}}]_*$. As such, we may iterate, with $<\kappa$-support, forcing notions of the form $\mathbb{LOC}_\kappa(B)$, using bookkeeping to ascertain that every $B \in \mathrm{BP}^{\lim}_\kappa$ (old or new) is used as a parameter cofinally often. By the following lemma, such iterations do not collapse cardinals.

Lemma 5.4. Let $\mathbb{P}_\mu = \langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha \mid \alpha \in \mu \rangle$ be a forcing iteration with $<\kappa$-support such that for each $\alpha \in \mu$ there is some $\mathbb{P}_\alpha$-name $\dot{B}^\alpha$ such that $\Vdash_{\mathbb{P}_\alpha}$ " $\dot{B}^\alpha \in \mathrm{BP}^{\mathrm{lim}}_\kappa$ and $\dot{\mathbb{Q}}_\alpha = \mathbb{LOC}_\kappa(\dot{B}^\alpha)$ ". Then $\mathbb{P}_\mu$ is strategically $<\kappa$-closed and $<\kappa^+$-c.c.

Proof. That strategic $<\kappa$-closure is preserved under $<\kappa$-support iteration is well-known (see [26, Proposition A.1.6] for a proof). For $<\kappa^+$-c.c., note that there is a dense set D of conditions $p \in \mathbb{P}_\mu$ such that for each $\alpha \in \mathrm{supp}(p)$ there exists $s_\alpha \in \mathbf{V}$ for which $p \restriction \alpha \Vdash$ " $\varphi_{p(\alpha)} \restriction \gamma_{p(\alpha)} = s_\alpha$ ". Any two conditions $p, p' \in \mathbb{LOC}_\kappa(B)$ with the same stem $\varphi_p \restriction \gamma_p = \varphi_{p'} \restriction \gamma_{p'}$ are compatible, hence we can apply the sunflower lemma (i.e., the Δ-system lemma). □

Theorem 5.5. Let $\mathbf{V} \vDash$ " GCH " and $\mu > \kappa$ be regular, then there exists a cardinal-preserving $<\kappa$-support iteration $\mathbb{P}_\mu$ of length μ, where each of the iterands is (forced to be) of the form $\mathbb{LOC}_\kappa(B)$, such that $\Vdash_{\mathbb{P}_\mu}$ " $\mathrm{non}(\mathcal{NA}_\kappa) = \mu$ ".

Proof. In the extension by $\mathbb{P}_\mu$, let $X \subseteq {}^\kappa 2$ with $|X| < \mu$ and let $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$. Then X has a $\mathbb{P}_\alpha$-name for some $\alpha < \mu$, and by bookkeeping there is some $\beta > \alpha$ such that $\mathbb{P}_\beta \Vdash$ " $\dot{\mathbb{Q}}_\beta = \mathbb{LOC}_\kappa(B)$ ". Then the $\dot{\mathbb{Q}}_\beta$-generic adds $\varphi \in \prod_{\alpha\in\kappa} \left[{}^{B_\alpha}2\right]^{|\alpha|}$ such that $X \subseteq ({}^\kappa 2)^{\mathbf{V}^{\mathbb{P}_\beta}} \subseteq [\varphi]_*$. Since B was arbitrary, we see that $X \in \mathcal{NA}_\kappa$. □

As a corollary (using Theorem 4.22), if κ is strongly inaccessible, then the above model satisfies

$$\forall B \in \mathrm{BP}^{\mathrm{lim}}_\kappa \left(\mathrm{add}(\mathcal{BE}^\forall_\kappa) = \mathrm{add}(\mathcal{E}^\forall_\kappa) = \mathrm{add}(\mathcal{E}_\kappa(B)) = \mu\right),$$

and for arbitrary regular uncountable κ, the model satisfies $\mathrm{non}(\mathcal{E}_\kappa(B)) = \mu$ for each $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$.

It is not hard to see that $\mathbb{LOC}_\kappa(B)$ adds a κ-Cohen generic; for instance, in $\mathbf{V}$, let $\prec_\alpha$ well-order ${}^{B_\alpha}2$ and define $c \in {}^\kappa 2$ by $c(\alpha) = 0$ if and only if $\mathrm{ot}(\varphi_{\mathrm{gen}}(\alpha), \prec_\alpha)$ is even, then c is κ-Cohen generic.

Lemma 5.6. If κ is weakly compact and $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$, then $\mathbb{LOC}_\kappa(B)$ is (κ, κ)-calibre.

Proof. Fix any condition $p^* \in \mathbb{LOC}_\kappa(B)$, let us write $s := \varphi_{p^*} \restriction \gamma_{p^*}$, then we claim that the set $C := \{p \in \mathbb{P} \mid \varphi_p \restriction \gamma_p = s\}$ has κ-calibre. We assume for convenience that $|B_\alpha| \geq \alpha$, the case for arbitrary $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$ is proved similarly.

Fix some sequence of distinct conditions $\langle p_\xi \mid \xi \in \kappa \rangle$ in C and note that $\gamma_{p_\xi} = \gamma_{p^*}$ for each ξ. Define a tree T where $t \in T$ if and only if there are $\xi \neq \xi'$ and $\gamma \geq \gamma_{p^*}$ with $t = \varphi_{p_\xi} \restriction \gamma = \varphi_{p_{\xi'}} \restriction \gamma$. Since κ is inaccessible, we see that $\left|\left\{\varphi \restriction \gamma \;\middle|\; \varphi \in \prod_{\alpha\in\kappa} \left[{}^{B_\alpha}2\right]^{|\alpha|}\right\}\right| < \kappa$ for each $\gamma \in \kappa$, and thus it follows that T is a $<\kappa$-branching tree of height κ. By weak compactness of κ, we then see that T has a branch $b \subseteq T$, and therefore $\psi := \bigcup b \in \prod_{\alpha\in\kappa} \left[{}^{B_\alpha}2\right]^{|\alpha|}$. Now, fix a strictly increasing sequence $\left\langle \xi(\gamma) \;\middle|\; \gamma \in [\gamma_{p^*}, \kappa) \right\rangle$ such that $\varphi_{p_{\xi(\gamma)}} \restriction \gamma = \psi \restriction \gamma$ for each γ, then if $\varphi' \colon \alpha \mapsto \bigcup_{\gamma\in[\gamma_{p^*},\kappa)} \varphi_{p_{\xi(\gamma)}}(\alpha)$ we see that $\varphi' \in \prod_{\alpha\in\kappa} \left[{}^{B_\alpha}2\right]^{|\alpha|}$ and $(\varphi', \gamma_{p^*}) \leq p_{\xi(\gamma)}$ for all $\gamma \in [\gamma_{p^*}, \kappa)$, as required. □

Corollary 5.7. If κ is weakly compact and $B \in \mathrm{BP}^{\mathrm{lim}}_\kappa$, then $\mathbb{LOC}_\kappa(B)$ does not add a dominating κ-real.

We do not know if the property of not adding dominating κ-reals is preserved in the iteration of Theorem 5.5.

5.1.2. *Adding a new slalom through a new partition.* The second forcing notion $\mathbb{LOC}^{\mathrm{new}}_\kappa$ is similar to $\mathbb{LOC}_\kappa(B)$, but it not only adds a new slalom φ, but also a new partition $I \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ for which $\varphi \in \Sigma_\kappa(I)$.

Definition 5.8. Every condition of $\mathbb{LOC}_\kappa(B)$ is a triple $p = (s_p, I^p, F_p)$ such that

(1) s_p and I^p are functions with $\mathrm{dom}(s_p) = \mathrm{dom}(I^p) \in \kappa$ and we write $I^p_\alpha := I^p(\alpha)$,

(2) if $\alpha \in \mathrm{dom}(s_p)$, then $I^p_\alpha \subseteq \kappa$ is a nonempty interval with $\min(\kappa \setminus \bigcup_{\xi<\alpha} I^p_\xi) = \min(I^p_\alpha)$ and $\sup(I^p_\alpha) < \kappa$,
(3) if $\alpha \in \mathrm{dom}(s_p)$, then $s_p(\alpha) \subseteq {}^{I^p_\alpha}2$ and $0 < |s_p(\alpha)| < 2^{|I^p_\alpha|}$,
(4) $F_p \subseteq {}^\kappa 2$ is a set of functions such that $|F_p| < \kappa$.

If $p, q \in \mathbb{LOC}^{\mathrm{new}}_\kappa$, then we let $q \leq p$ if and only if $s_p \subseteq s_q$ and $F_p \subseteq F_q$, and for any $\alpha \in \mathrm{dom}(s_q \setminus s_p)$ and $f \in F_p$ we have $f \restriction I^q_\alpha \in s_q(\alpha)$.

If $\mathbf{V}$ is a model of ZFC and G is $\mathbb{LOC}^{\mathrm{new}}_\kappa$-generic over $\mathbf{V}$, then $I^{\mathrm{gen}} := \bigcup_{p\in G} I^p \in \mathrm{IP}^{\mathrm{lim}}_\kappa$ and $\varphi_{\mathrm{gen}} := \bigcup_{p\in G} s_p \in \Sigma_\kappa(I^{\mathrm{gen}})$. By genericity, $(2^\kappa)^{\mathbf{V}} \subseteq [\varphi_{\mathrm{gen}}]_*$. Compared to $\mathbb{LOC}_\kappa(B)$, the forcing notion $\mathbb{LOC}^{\mathrm{new}}_\kappa$ has stronger closure and linkedness properties:

Lemma 5.9. $\mathbb{LOC}^{\mathrm{new}}_\kappa$ is $<\kappa$-closed and $(\kappa, <\kappa)$-centred.

Lemma 5.10. Let $\mathbb{P}_\mu = \langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha \mid \alpha \in \mu\rangle$ be a forcing iteration with $<\kappa$-support such that for each $\alpha \in \mu$ we have $\Vdash_{\mathbb{P}_\alpha}$ "$\dot{\mathbb{Q}}_\alpha = \mathbb{LOC}^{\mathrm{new}}_\kappa$". Then $\mathbb{P}_\mu$ is $<\kappa$-closed, and if $\mu < (2^\kappa)^+$, then $\mathbb{P}_\mu$ is $(\kappa, <\kappa)$-centred.

Proof. The preservation of $(\kappa, <\kappa)$-centredness follows as in [8, Lemma 55]. □

Theorem 5.11. Let $\mathbf{V} \vDash$ "GCH" and let $\mu > \kappa$ be regular. Let $\mathbb{C}$ be the μ-length $<\kappa$-support product of κ-Cohen forcing and let $\dot{\mathbb{P}}_\mu$ be a $\mathbb{C}$-name for the μ-length $<\kappa$-support iteration such that for each $\alpha \in \mu$ we have $\Vdash_{\mathbb{C}*\dot{\mathbb{P}}_\alpha}$ "$\dot{\mathbb{Q}}_\alpha = \mathbb{LOC}^{\mathrm{new}}_\kappa$". Then

$$\Vdash_{\mathbb{C}*\dot{\mathbb{P}}_\mu} \text{"}\forall B \in \mathrm{BP}^{\mathrm{lim}}_\kappa(\kappa^+ = \mathrm{non}(\mathcal{E}_\kappa(B)) < \mathrm{cov}(\mathcal{M}_\kappa) = \mathrm{non}(\mathcal{E}_\kappa) = \mu = 2^\kappa)\text{"}.$$

Proof. Let H be $\mathbb{C}$-generic and note that $\mathbf{V}[H] \vDash$ "$\kappa^+ = \mathrm{non}(\mathcal{M}_\kappa) < \mathrm{cov}(\mathcal{M}_\kappa) = \mu = 2^\kappa$". It then follows from Lemma 5.10 that $\mathbb{P}_\mu := (\dot{\mathbb{P}}_\mu)^{[H]}$ is $(\kappa, <\kappa)$-centred in $\mathbf{V}[H]$. [31, Definition 65] defines a cardinal characteristic $\mathfrak{sup}_\kappa(\not\in^\infty)$, and from [31, Lemma 79] we know that $(\kappa, <\kappa)$-centred forcing notions preserve $\mathfrak{sup}_\kappa(\not\in^\infty) = \kappa^+$. Since $\mathfrak{b}^{b,b}_\kappa(\in^*) \leq \mathfrak{sup}_\kappa(\not\in^\infty) \leq \mathrm{non}(\mathcal{M}_\kappa)$, we may use Theorem 4.17 to conclude that $\Vdash_{\mathbb{C}*\dot{\mathbb{P}}_\mu}$ "$\forall B \in \mathrm{BP}^{\mathrm{lim}}_\kappa(\mathrm{non}(\mathcal{E}_\kappa(B)) = \kappa^+)$".

Now, working in $\mathbf{V}[H]$, suppose that $\dot{X}$ is $\mathbb{P}_\mu$-name for a subset of ${}^\kappa 2$ of size $<\mu$, then there is a stage $\gamma < \mu$ such that $\dot{X}$ is a $\mathbb{P}_\gamma$-name. But, as we discussed above, in the next stage of the iteration a slalom φ_{gen} is added such that $\dot{X} \subseteq [\varphi_{\mathrm{gen}}]_*$, thus $\Vdash_{\mathbb{P}_\mu}$ "$\dot{X} \in \mathcal{E}^\exists_\kappa$". Consequently, $\Vdash_{\mathbb{C}*\dot{\mathbb{P}}_\mu}$ "$\mathrm{non}(\mathcal{E}^\exists_\kappa) = \mu$" and thus the theorem follows from $\mathrm{non}(\mathcal{E}^\exists_\kappa) \leq \mathrm{non}(\mathcal{E}_\kappa)$. □

It is again easy to see that $\mathbb{LOC}^{\mathrm{new}}_\kappa$ adds a κ-Cohen generic. Moreover, if we assume that κ is weakly compact, then $\mathbb{LOC}^{\mathrm{new}}_\kappa$ does not add dominating κ-reals, but we do not know how to preserve this property under iteration, thus we will omit the details.

5.2. **The Laver property.** We make a remark about the Laver property. Classically, a forcing notion with the Laver property can be iterated with countable support to obtain a model in which $\mathrm{cov}(\mathcal{E}_\kappa)$ is small, see for istance [3, Lemma 7.2.3]. The same argument works in the higher context as well. One well-known higher forcing notion that has a higher Laver property, is the κ-Sacks forcing of [20] as well as its iteration under $\leq\kappa$-support, see also [8, § 5.1] and [23, Proposition 1.86]. We may therefore conclude that if κ is strongly inaccessible and we assume GCH, then the κ^{++}-length iteration with $\leq\kappa$-support of the κ-Sacks forcing will result in a model where $\mathrm{cov}(\mathcal{E}_\kappa) = \kappa^+ < 2^\kappa$, and moreover, as shown in [8], the resulting model additionally satisfies $\mathrm{cof}(\mathcal{M}_\kappa) = \kappa^+$.

References

[1] Claudio Agostini, Fernando Barrera, and Vincenzo Dimonte. On the problem of generalized measures: an impossibility result, 2026. Preprint.

[2] Tomek Bartoszyński. Combinatorial aspects of measure and category. *Fundam. Math.*, 127(3): 225–239, 1987. doi: 10.4064/fm-127-3-225-239.

[3] Tomek Bartoszyński and Haim Judah. *Set Theory: On the Structure of the Real Line*. A.K. Peters, Wellesley, MA, 1995.

[4] Tomek Bartoszyński and Saharon Shelah. Closed measure zero sets. 58(2):93–110. doi: 10.1016/0168-0072(92)90001-G. URL https://linkinghub.elsevier.com/retrieve/pii/016800729290001G.

[5] Tom Benhamou and Corey Bacal Switzer. Eventual capture on a measurable cardinal, 2025. Preprint.

[6] A. Blass. Cardinal characteristics and the product of countably many infinite cyclic groups. *J. Algebra*, 169(2):512–540, 1994. doi: 10.1006/jabr.1994.1295. URL http://dx.doi.org/10.1006/jabr.1994.1295.

[7] Andreas Blass, Tapani Hyttinen, and Yi Zhang. Mad families and their neighbors. URL https://dept.math.lsa.umich.edu/~ablass/madfam.pdf. Preprint.

[8] Jörg Brendle, Andrew Brooke-Taylor, Sy-David Friedman, and Diana Carolina Montoya. Cichoń's diagram for uncountable cardinals. *Isr. J. Math.*, 225(2):959–1010, 2018. doi: 10.1007/s11856-018-1688-y.

[9] Jörg Brendle. Evasion and prediction - the specker phenomenon and gross spaces. *Forum Math.*, 7: 513–542, 1995. doi: doi:10.1515/form.1995.7.513. URL https://doi.org/10.1515/form.1995.7.513.

[10] Miguel A. Cardona, Adam Marton, and Jaroslav Supina. Cardinal characteristics associated with small subsets of reals, 2025. Preprint.

[11] Nick Steven Chapman and Johannes Philipp Schürz. Strong measure zero sets on 2^κ for κ inaccessible. *J. Symb. Log.*, 90(3):1277–1307, 2024. doi: 10.1017/jsl.2023.100. URL http://dx.doi.org/10.1017/jsl.2023.100.

[12] Shani Cohen and Saharon Shelah. Generalizing random real forcing for inaccessible cardinals. *Isr. J. Math.*, 234:547–580, 2019. doi: 10.1007/s11856-019-1925-z.

[13] James Cummings and Saharon Shelah. Cardinal invariants above the continuum. *Ann. Pure Appl. Log.*, 75:251–268, 1995. doi: 10.1016/0168-0072(95)00003-Y.

[14] Sy-David Friedman and Giorgio Laguzzi. A null ideal for inaccessibles. *Arch. Math. Log.*, 56:691–697, 2017. doi: 10.1007/s00153-017-0562-7.

[15] Fred Galvin, Jan Mycielski, and Robert M. Solovay. Strong measure zero sets. *Not. Am. Math. Soc.*, 26:p. A–280, 1979.

[16] Fred Galvin, Jan Mycielski, and Robert M. Solovay. Strong measure zero and infinite games. *Arch. Math. Log.*, 56(7–8):725–732, 2017. doi: 10.1007/s00153-017-0541-z. URL http://dx.doi.org/10.1007/s00153-017-0541-z.

[17] Martin Goldstern and Saharon Shelah. Many simple cardinal invariants. *Arch. Math. Log.*, 32(3): 203–221, 1993. doi: 10.1007/BF01375552.

[18] Aapo Halko. *Negligible subsets of the generalized Baire space* $\omega_1^{\omega_1}$. The Finnish Mathematical Society, 1996. doi: 10.5186/aasfmd.1996.107. URL http://dx.doi.org/10.5186/aasfmd.1996.107.

[19] Aapo Halko and Saharon Shelah. On strong measure zero subsets of ${}^\kappa 2$. *Fundam. Math.*, 170(3): 219–229, 2001. doi: 10.4064/fm170-3-1. URL http://dx.doi.org/10.4064/fm170-3-1.

[20] Akihiro Kanamori. Perfect-set forcing for uncountable cardinals. *Annals of Mathematical Logic*, 19 (1-2):97–114, 1980. doi: 10.1016/0003-4843(80)90021-2.

[21] Jakob Kellner. Even more simple cardinal invariants. *Arch. Math. Log.*, 47(5):503–515, 2008. doi: 10.1007/s00153-008-0094-2.

[22] Arnold W. Miller. Some properties of measure and category. *Trans. Am. Math. Soc.*, 266(1):93–114, 1981. doi: 10.1090/S0002-9947-1981-0613787-2.

[23] Diana Carolina Montoya Amaya. *Some cardinal invariants of the generalized Baire spaces*. Phd thesis, Universität Wien, 2017.

[24] Dilip Raghavan and Saharon Shelah. Two inequalities between cardinal invariants. *Fundam. Math.*, 237(2):187–200, 2017. doi: 10.4064/fm253-7-2016. URL http://dx.doi.org/10.4064/fm253-7-2016.

[25] Andrzej Roslanowski. żaba: a case for clubs, 2026. Preprint.

[26] Andrzej Rosłanowski and Saharon Shelah. Sheva-sheva-sheva: Large creatures. *Israel Journal of Mathematics*, 159(1):109–174, 2007. doi: 10.1007/s11856-007-0040-8.

[27] Johannes Philipp Schürz. The cofinality of the strong measure zero ideal for κ inaccessible. *Math. Log. Q.*, 69(1):31–39, 2023. doi: 10.1002/malq.202000093. URL http://dx.doi.org/10.1002/malq.202000093.

[28] Saharon Shelah. Every null-additive set is meager-additive. *Isr. J. Math.*, 89(1–3):357–376, 1995. doi: 10.1007/bf02808209. URL http://dx.doi.org/10.1007/BF02808209.

[29] Saharon Shelah. A parallel to the null ideal for inaccessible λ: Part I. *Arch. Math. Log.*, 56(3-4):319–383, 2017. doi: 10.1007/s00153-017-0524-0. URL https://doi.org/10.1007/s00153-017-0524-0.

[30] Toshio Suzuki. About splitting numbers. *Proc. Jpn. Acad., Ser. A, Math. Sci.*, 74(2), 1998. doi: 10.3792/pjaa.74.33. URL http://dx.doi.org/10.3792/pjaa.74.33.

[31] Tristan van der Vlugt. Cardinal characteristics on bounded generalised Baire spaces. *Ann. Pure Appl. Log.*, 176(7):103582, 2025. doi: 10.1016/j.apal.2025.103582. URL http://dx.doi.org/10.1016/j.apal.2025.103582.

[32] Wolfgang Wohofsky. *Special sets of real numbers and variants of the Borel Conjecture*. PhD thesis, TU Wien, 2013.

[33] Ned Wontner. Views from a peak: Generalisations and descriptive set theory. *Doctoral dissertation, University of Amsterdam*, 2023.

[34] Jindřich Zapletal. Splitting number at uncountable cardinals. *J. Symb. Log.*, 62(1):35–42, 1997. doi: 10.2307/2275730.

[35] Ondřej Zindulka. Strong measure zero and meager-additive sets through the prism of fractal measures. *Commentationes Math. Univ. Carolinae*, 60(1):131–155, 2019. doi: 10.14712/1213-7243.2015.277. URL http://dx.doi.org/10.14712/1213-7243.2015.277.